



Integrated optimization of train timetable and train unit circulation for a Y-type urban rail transit system with flexible train composition mode[☆]

Lin Yang^a, Yuan Gao^{a,*}, Andrea D'Ariano^b, Suxiu Xu^a

^a School of Management and Economics, Beijing Institute of Technology, Beijing 100081, China

^b Department of Civil, Computer Science and Aeronautical Technologies Engineering, Roma Tre University, Rome 00146, Italy

ARTICLE INFO

Keywords:

Train operation plan
Flexible train composition mode
Integer programming
Adaptive large neighborhood search

ABSTRACT

With the increasing complexity of urban rail transit (URT) networks and the revolution of train control technology, a large number of diverse operation and organization modes have emerged. This paper studies the train operation plan problem for a Y-type URT system with flexible train composition mode. The studied problem can be defined as an integrated optimization of the train timetable problem (TTP) and train unit circulation problem (TUCP) under time-dependent passengers, where the train compositions can be coupled or decoupled during the trip. By considering train operation constraints, train composition constraints, train unit circulation constraints, and passenger demand constraints with limited train capacity, our problem is formulated as an integer programming (IP) model, where the objective is to minimize the passengers' total waiting time and the number of train units used. To solve large-scale real-world instances, we design a heuristic algorithm by improving the classical framework of the adaptive large neighborhood search (ALNS) algorithm. Several groups of experiments, including small-case instances and real-world instances of Guangzhou Metro Line 14, are investigated to verify the effectiveness of the proposed model and algorithm. Compared with the traditional fixed train composition mode, both the passengers' total waiting time and the number of train units used under the condition of flexible train composition mode are reduced. This will effectively improve the quality of the passenger service and reduce the costs of URT departments.

1. Introduction

In an urban rail transit (URT) system, a train runs in the upstream/downstream direction and stops at some stations for passengers' boarding and alighting, which is referred to as an upstream/downstream service. In the most common operation mode, homogeneous services with the same train unit (i.e., definite carriages) run along a URT line in downstream and upstream directions. However, this mode leads to the transfer activity of passengers who commute on different lines. Some special URT lines/networks have been designed to avoid this inconvenience. For example, in the Y-type system, a branch extends outward from the middle of a URT line. Typically, a Y-type URT system consists of three sub-lines: *trunk line*, *major siding line*, and *minor siding line* (see Fig. 1). Two types of routes are operated, which are defined as *major route* and *minor route*. Passengers usually board the services with the route matching their travel paths.

In addition to the improvement of the topological structure of URT lines, there has also been a revolution in train control systems. The virtual coupling technology [1,2] based on train-to-train (T2T) communication has been proposed. This makes train units flexible enough to

couple/decouple during trips. An operation mode called *flexible train composition* then appears [3–5]. The key feature of the flexible train composition mode is that train compositions can be formed through coupling/decoupling operations of some train units during trips, which provides different capacities for passengers on different segments. Obviously, in the flexible train composition mode, the capacity of train compositions can better match the passenger demand. In comparison, the train composition of each service is fixed in the traditional *fixed train composition* mode and cannot be changed for different passenger demands.

In a Y-type system, passenger demand on the trunk line and major siding line is usually larger than on the minor siding line. The flexible train composition mode for a Y-type system is used to balance the quality of passenger service and the supply of transportation capacity. In our problem, we consider two types of train compositions running on a Y-type URT system; that is, the *single-composition* with one train unit and the *double-composition* with two train units (see Fig. 1). Note that these two types of train compositions can be converted into each

[☆] Area: Production Management, Scheduling and Logistics. This manuscript was processed by Associate Editor Yagiura.

* Corresponding author.

E-mail address: gaoyuan@bit.edu.cn (Y. Gao).

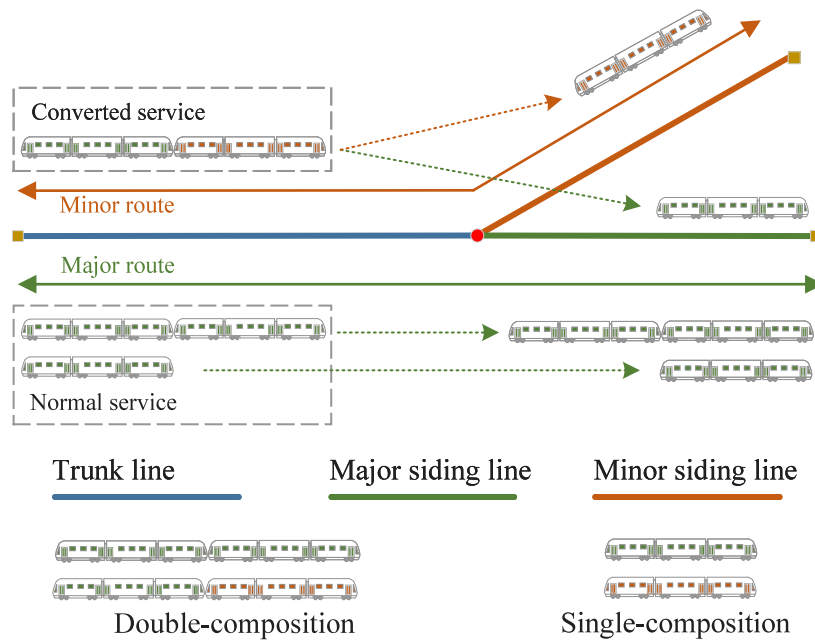


Fig. 1. An illustration of train compositions and routes.

other by coupling/decoupling, where the converting operation can be implemented during trips. In view of this, it is possible to let some double-composition trains from the trunk line be decoupled into two single-composition trains. One of them then goes to the major siding line and the other goes to the minor siding line, and vice versa for the opposite direction by coupling operation. It should be noted that both services for the major siding line and the other for the minor siding line before coupling (after decoupling) are operated under one train number. For convenience, the services with coupling/decoupling operations during the trip are named *converted services*, otherwise as *normal services* (see Fig. 1). A practical application of the concept can refer to the S-Bahn Linie S1 in Munich, Germany, and Kansai Airport/Kishiyuji Rapid, Japan.

Although the flexible train composition mode in the Y-type URT system can improve the service quality and the matching degree between transportation capacity and passenger demand, it can also bring complexity to the train operation plan; that is, *Train timetable problem* (TTP) and *train unit circulation problem* (TUCP). Generally, TTP is the premise of TUCP, and they are optimized step by step. However, considering the location of depots and the change of train composition in a Y-type system, a pre-given timetable may not satisfy the constraints of a train unit circulation plan. For example, the first service starting from a turnaround station without depots cannot be implemented until an opposite service arrives at the station. Therefore, we think that it is more meaningful to integrate the optimization of the train timetable and train unit circulation. The purpose of this study is to develop an integrated optimization method of TTP and TUCP by considering the flexible train composition mode for a Y-type system to achieve a high-quality operation plan. Through this method, the passenger demand can be better matched with the planned transportation capacity. The investment cost of the train units can then be further reduced from the perspective of the system.

To address this problem, we first formulate it as a mathematical programming model, with an objective to minimize the passengers' total waiting time and the number of train units used, where they are handled by the linear weighted method. Note that the running times in segments are fixed, thus the change of each passenger's in-vehicle time is slight, which is only caused by the coupling/decoupling of services. In view of this, we choose the waiting time of passengers as one of our objective functions. Furthermore, considering the requirements of less

computation time for URT departments and the inability of commercial solvers to solve the proposed model in large-scale instances, we improve the classical framework of the adaptive large neighborhood search (ALNS) algorithm by introducing asynchronous computation mechanisms and we design a heuristic algorithm to solve our model. Finally, we generate some computational experiments to evaluate our methodology and to give insights to the URT departments, from which practitioners can quickly obtain some decision-making information—including a train timetable and a train unit circulation plan—to match the passenger demand and the supply of transportation capacity in different scenarios.

1.1. Literature review

The train operation plan problem in the URT system consists of several parts, including network plan, line plan, train timetable and train unit circulation [6]. Recalling our research problems, the network plan and line plan are pre-given in advance, which are a Y-type URT system and two types of routes. Therefore, we focus on the literature related to TTP and TUCP, in which TTP determines the arrival and departure time of services at stations, and TUCP specifies the services performed by train units (including the sequence plan and assignment plan of train units) [7].

1.1.1. Train timetable problem (TTP)

As the key point of a train operation plan in railway system, TTP is used to produce a conflict-free schedule for the operations of services. This directly affects the safety and efficiency of the practical operation. There are two main areas in the TTP, namely at an *operational level* (dispatching or rescheduling) and at a *planning level* (macroscopic timetable scheduling). Cacchiani et al. [8] gave a detailed overview of early research about the first area. After that, some related studies have been developed for some special problems in algorithms and models, such as decomposition approaches [9–12], multi-criteria decision rescheduling model at a complex railway network [13], rescheduling model with considering uncertain factors [14,15], and the dynamic rescheduling problem [16–18]. For TTP at the planning level, the key problem is how to design high-quality timetables for given operational parameters (i.e., running times in segments, safety headway, and dwelling times) and train stop patterns. In general, operation parameters are relatively

stable and are derived from past operating experience or the standards in the network plan. Thus, the stop patterns play an important role when making a train timetable.

The train stop patterns are commonly classified as all-stop operation, skip-stop operation, zonal operation and express/local operation [19]. In a URT system, the all-stop operation is more widely applied, while others have increasingly been used in recent years. Niu and Zhou [20] studied the optimization of a passenger train timetable in the heavily congested URT corridor, where all services run with an all-stop operation. Later, they established a nonlinear integer programming model with linear constraints to minimize the total waiting time of passengers by inputting the pre-given skip-stop operation [21]. Furthermore, some researchers have opted to introduce train stop status as a set of decision variables to determine the timetable and stopping patterns of a set of services simultaneously [22–26]. In an oversaturated URT system, combined with rigid first-in-first-out rule and capacity constraints, Wu et al. [27] optimized the train skip-stop mode and adjusted the train timetable to alleviate the unfairness of passenger waiting time. Tang et al. [28] studied the train timetabling problem under the express/local stopping mode, in which express services can skip certain low-demand stations.

1.1.2. Train unit circulation problem (TUCP)

TUCP is also known as the *train unit scheduling problem* or *rolling stock circulation/scheduling problem*. Generally, TUCP determines how to make rational use of the train unit resources in the system within a specific time horizon. To solve this problem, some premises are set in advance (e.g., the existing timetable, pre-defined passenger demand, and limited types of train units). Characterized by whether there are coupling/decoupling operations, two train composition modes are defined; that is, flexible train compositions [3–5,29,30] and fixed train compositions [27,31,32].

Formulating an optimization model for a practical TUCP is a common method. At an early stage, based on the pre-given timetable and passenger demand, Schrijver [29] established an optimization model to minimize the number of two types of train units. The proposed frame based on spatio-temporal graphs became an important modeling basis. Abbink et al. [33] proposed a model to find an optimal allocation of train types and subtypes to the lines of Dutch railway. The results solved by CPLEX were better than the manually planned result. Steinzen et al. [31] formulated an optimization model based on spatio-temporal network, which was solved by column generation in combination with Lagrangian relaxation. Canca and Barrena [34] proposed a general mixed integer programming model to design the rolling stock circulation plan, which simultaneously determines the number and location of rest facilities. Moreover, TUCP has also been turned into an integer multi-commodity flow model [35,36], where the commodities correspond to the rolling stock subtypes and the graph is determined by the different services that must be served. By considering several objective standards related to operation cost, service quality and railway system reliability, Fioole et al. [37] extended the existing rolling stock model [36] and verified the result in a real-life instance.

As an exact algorithm, the branch-and-price (BP) algorithm is widely used to solve TUCP [7,30,36,38]. Several heuristic algorithms have also been applied to TUCP to quickly obtain a satisfactory feasible solution. Cacchiani et al. [39] designed the LP-based heuristic method to solve an integer linear programming model, which is characterized by a considerable number of different train unit types and services. Cacchiani et al. [40] proposed a fast Lagrangian heuristic algorithm for TUCP, which calls for the definition of train units to be assigned to a given set of services and each service requires a given number of passenger seats. The computational results show that the proposed heuristic method is faster than the existing methods in practice. Considering the practical requirements of a URT system, Thorlacius et al. [41] proposed a comprehensive rolling stock plan model that was used to improve the existing rolling stock plan by using the mountain

climbing heuristic method. Under the condition of a pre-given set of train units and a set of services with passenger seat demand and timetables, Cacchiani et al. [32] established an effective heuristic for TUCP at peak-period to achieve the matching problem from train units to services with the objective of minimizing the cost. When the services in a timetable are canceled due to the interruption, the rolling stock plan needs to be rescheduled. A variable neighborhood search heuristic algorithm for the rolling stock rescheduling problem has been designed to improve passenger comfort and operation performance [42].

1.1.3. Integrated optimization of TTP and TUCP

Research on integrated optimization of TTP and TUCP in URT systems has been carried out with both fixed and flexible train composition modes. In terms of fixed train composition mode, Yue et al. (2017) proposed a bi-level programming model that optimizes train timetable and rolling stock scheduling based on time-dependent passenger demand, balancing the trade-off between waiting time and service frequency. Similarly, Wang et al. [43] developed an integrated model to optimize train timetable and train unit circulation plan, taking into account various operations such as service operation, turnaround operation, and depot operation, among others. For peak-period URT systems with overcrowded and unevenly-distributed passenger demand, Yuan et al. [44] introduced a new optimization model of timetable, rolling stock assignment, and short-turning strategy to minimize total passenger waiting time. Wang et al. [45] proposed a time-space network representation and a row and column generation-based algorithm to optimize rolling stock routing and train timetable. Wang et al. [46] formulated a complex multi-objective MILP model to reschedule trains and rolling stock circulation for complete blockage situations in metro lines, including deviations with respect to the timetable, cancellations, and headway deviations of services. Liu et al. [47] developed a collaborative optimization model for train timetable and train unit turnarounds in metro lines, incorporating passenger control strategy and using a Lagrange relaxation method.

In terms of flexible train composition mode, Cadarso and Marín [48] aimed to formulate a unified model for railway transportation networks by integrating the planning phases of train scheduling and rolling stock material and composition. To address this problem, they employed a non-cyclic approach and considered various factors such as limited rolling stock resources, station capacities, and passenger demands. The model also took into account shunting operations, including empty movements, composition changes, and rotations, with an emphasis on enhancing robustness by penalizing hazardous shunting operations and enabling swapping operations at strategic depot stations. In a subsequent study, Cadarso et al. [49] further addressed the timetabling and rolling stock scheduling problem by proposing a mixed integer linear programming model, which was presented as the second step in a two-step method for railway disruption management. It is worth noting that the model proposed by Cadarso et al. [49] included passenger demand behavior and the strategy of inserting emergency services. In recent years, the concept of flexible train composition has been widely mentioned. Pan et al. [3] presented a column-generation-based diving heuristic algorithm to optimize train timetable and rolling stock circulation plan with flexible train composition mode for URT lines. Zhou et al. [4] proposed a MILP model to optimize train timetable and rolling stock circulation plan under tidal passenger demand, allowing rolling stock to change their compositions through coupling/decoupling operations at both ends of URT lines. Ying et al. [5] used a multi-agent deep reinforcement learning approach to develop an adaptive control system for coordinated metro operations with the flexible train composition mode. By considering flexible train formation, Zhao et al. [50] proposed a multi-objective mixed-integer nonlinear programming model to integrate train timetable and rolling stock circulation for an urban rail transit line, with the aims of minimizing costs and meeting passenger demand.

Table 1
Summary of relevant studies.

Publications	Premise	Passengers	Integration	Train compositions	Constraints ^a	Objectives ^b	Solution methods
Wu et al. [27]	Line	Time-dependent	TTP	Fixed	Capacity+FIFO	MWT↓	VNS+CPLEX
Yang et al. [51]	Y-type system	Macro	TTP	Fixed	–	TTS↑, OCS↑	GA
Lusby et al. [30]	Train services	–	TUCP	Flexible	–	NoTU↓, SC↓	Branch and price
Cacchiani et al. [32]	Train services	Macro	TUCP	Fixed	Capacity	NoTU↓	Heuristic
Cadarso and Marín [48]	Network	Period-dependent	TTP, TUCP	Flexible	Capacity	TC↓, TP↓	CPLEX
Cadarso et al. [49]	Network	Time-dependent	TTP, TUCP	Flexible	Capacity	OC↓, CC↓, SC↓, DP↓, PD↓	CPLEX
Wang et al. [43]	Line	–	TTP, TUCP	Fixed	–	HD↓, HV↓	CPLEX
Wang et al. [46]	Line	–	TTP, TUCP	Fixed	–	HD↓, HV↓, SC↓	Two-stage approach
Wang et al. [52]	Line	–	TTP, TUCP	Fixed	–	HD↓, HV↓, EC↓	CPLEX
Liu et al. [47]	Line	Station-dependent	TTP, TUCP	Fixed	–	NoGP↓	Lagrangian relaxation
Yuan et al. [44]	Line	Time-dependent	TTP, TUCP	Fixed	Capacity	WT_R↓	GA
Pan et al. [3]	Line	Time-dependent	TTP, TUCP	Flexible	Capacity	TC↓, WT↓	CG-based heuristic
Ying et al. [5]	Line	Time-dependent	TTP, TUCP	Flexible	Capacity	ORS↑	MADDPG
Zhou et al. [4]	Line	Time-dependent	TTP, TUCP	Flexible	Capacity	TOC↓, TWT↓	VNS+CPLEX
This paper	Y-type system	Time-dependent	TTP, TUCP	Flexible	Capacity+FIFO	TWT↓, NoTU↓	ALNS

^a In this column, we compare whether two types of constraints are included in these Publications, wherein, *Capacity* indicates the train capacity constraint, and *FIFO* indicates the constraints associated with the passenger boarding rule of first in first out.

^b The symbols denote: ↓ = Minimize; ↑ = Maximize; MWT = Maximum waiting time; TTS = Travel time savings; OCS = Operating cost savings; NoTU = Number of train units; SC = Services cancellations; TC = Total cost; TP = Total penalty; OC = Operating costs; CC = Composition changes; DP = Denied passengers; PD = Penalty for deviations; HD = Headway deviations; HV = Headway variations; EC = Energy consumption; NoGP = Number of gathered passengers; WT_R = Waiting time calculated based on Riemann upper sum; ORS = Overall reward signals; TOC = Total operation cost; TWT = Total waiting time.

1.2. Contributions

From this review of the literature, it can be seen that there has been much research on TTP and TUCP. To illustrate the innovation of this research, Table 1 compares the studies that are closely related to our problem or method from three aspects: problem boundaries (premise, types of passenger demand, integration of different problems, and train compositions), mathematical programming (capacity constraints, boarding rule constraints, and objectives) and solution methods. Some conclusions follow:

- (1) Most studies considered a line rather than a complicated topological structure, except for Yang et al. [51], Cadarso and Marín [48] and Cadarso et al. [49]. Among them, Yang et al. [51] considered a Y-type system similar to our problem, but this work only included TTP. Cadarso and Marín [48] and Cadarso et al. [49] considered the integration of TTP and TUCP for railway networks consisting of multiple lines, but the passenger demand involved was based on groups and each group of passengers was not allowed to be split. Further, none of these three studies considered the passenger boarding rule of first in first out (FIFO).
- (2) In terms of passengers, waiting time—including maximum waiting time [27] and total waiting time [3,44]—is always considered to be the most important factor. On the other hand, operators pay more attention to operating cost, such as operating cost saving [49,51], energy consumption [52], number of train units used [30,32], and total involved cost [3,48]. In this paper, to balance the service quality and operating cost, we choose the total waiting time and the number of train units as our optimization objectives.
- (3) When both the time-dependent passengers and train capacity constraint are considered, the computational complexity of the problem increases significantly and it becomes difficult to solve a large-scale problem to optimality by using exact algorithms. Thus, some heuristic algorithms are adopted to obtain a near optimal solution [3,27,44].

With these concerns, the key contributions of this paper are as follows:

- (1) Although the integrations of TTP and TUCP for railway networks were conducted by Cadarso and Marín [48] and Cadarso et al. [49], our work is distinct from theirs since we consider more accurate passenger demand and the realistic FIFO boarding rule. Specifically, we investigate the integrated TTP and

TUCP problem for a Y-type URT system with a flexible train composition mode. We also address the challenge of matching train unit routes with passenger travel paths. To address these issues, we propose a mathematical programming model aimed at minimizing passenger waiting time and the number of train units required. As far as we know, our study is the first to tackle the specific combination of these problem characteristics.

- (2) Based on the decomposable features of our model, we design a heuristic algorithm by improving the classical framework of the adaptive large neighborhood search (ALNS) algorithm to solve the proposed problem. In particular, the solution framework is divided into two levels—the asynchronous level and the global level. The asynchronous level is used to search a group of new solutions in parallel by executing operators. These generated solutions and the selected operators are recorded and returned to the global level, where the parameters and solutions are updated according to the searching strategy and the solutions obtained at the asynchronous level. We also develop a new methodology to identify feasible solutions efficiently for our problem.
- (3) Computational experiments on a group of small-case instances and large-scale real-world instances are performed, which show the effectiveness and efficiency of our method to obtain a near optimal solution. By comparing with the fixed train composition mode, it is proved that our solution to the problem that we studied has obvious advantages in reducing passenger waiting time and the number of train units used.

The remainder of this paper is organized as follows. Section 2 describes the problems, including the Y-type system, train unit routes and train unit circulation, time-dependent passengers, and boarding rules. A mathematical programming model considering an integrated train operation plan is constructed in Section 3. Section 4 introduces a heuristic algorithm to solve the proposed problem and Section 5 presents several groups of computational experiments. Section 6 concludes this paper and our works and presents some directions for future research.

2. Problem statement

2.1. Network structure

A bidirectional double-track Y-type URT system consists of three parts: trunk line, major siding line and minor siding line. We define both the major siding line and minor siding line as *siding lines* to

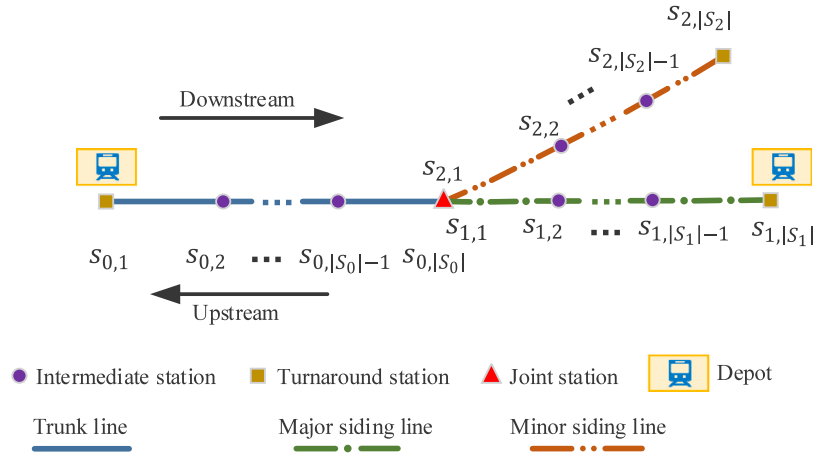


Fig. 2. The structure of a Y-type URT system.

distinguish them from the trunk line. Let L denote the set of lines and each line is indexed by l ($l \in L$). A line l contains a set of stations $(s_{l,1}, s_{l,2}, \dots, s_{l,|S_l|})$, where S_l denotes the set of stations on line l and $|S_l|$ represents the number of stations on line l . We further define the set of stations in the Y-type URT system $S = \bigcup_{l \in L} S_l$. Note that the first station of each siding line is actually the last station of the trunk line. We then define them as *joint stations*, denoted as S^{js} . Moreover, the first station of the trunk line and the last station of each siding line are *turnaround stations*, and the set of turnaround stations is denoted as S^{ts} . The *depots* near the turnaround stations provide train units for the train compositions used in the URT system. The other stations are called *intermediate stations*, where all train compositions stop for the passengers to board or alight. Furthermore, the direction from the trunk line to the siding lines is defined as *downstream direction*, otherwise as the *upstream direction*. For the convenience of modeling, it is specified that the stations on each line are numbered in consecutive integers along the downstream direction. In a line l ($l \in L$), the section between station s and $s+1$ is defined as *segment s* ($s \in S_l \setminus \{s_{l,|S_l|}\}$). As shown in Fig. 2, trunk line, major siding line, and minor siding line are marked as line 0, line 1, and line 2, respectively, and two depots are located near station $s_{0,1}$ and station $s_{1,|S_1|}$, separately. If we number the first station of the trunk line as station 1, then $S_0 = \{1, 2, \dots, |S_0|\}$, and the first station of the major siding line and minor siding line is numbered as $|S_0| + 1$ and $|S_0| + |S_1| + 1$, respectively.

2.2. Train unit routes and train unit circulation

Recall that there are two types of routes in the Y-type URT system; that is, the major route and the minor route (see Fig. 1). Denote the set of routes as $R = \{1, 2\}$ (indexed by r), where $r = 1$ and $r = 2$ represent the major route and minor route, respectively. Generally, the passenger demand on the minor siding line is small. Take Shanghai Metro Line 10 as an example. The headway of the minor siding line (Hangzhong road - Longxi road) is about 12 minutes, while that of the major siding line (Hongqiao railway station - Longxi road) and trunk line (Longxi road - Jilong road) is about 6 minutes and 4 minutes, respectively. For an integrated problem of TTP and TUCP, it is possible to cancel or insert some services to obtain more appropriate planning [30,46,48,49]. In our problem, we set the service frequency of the minor siding line to unfixed. That is to say, we still consider services with these two routes. Unlike the fixed train composition mode, we assume that the minor route on the part of the siding line is always operated by a single-composition, which is decoupled from a double-composition of a converted service at the joint station in the downstream direction, and vice versa for coupling operation in the upstream direction.

At the turnaround stations with depots, the train units of a starting service may come from different sources (i.e., the depots or train

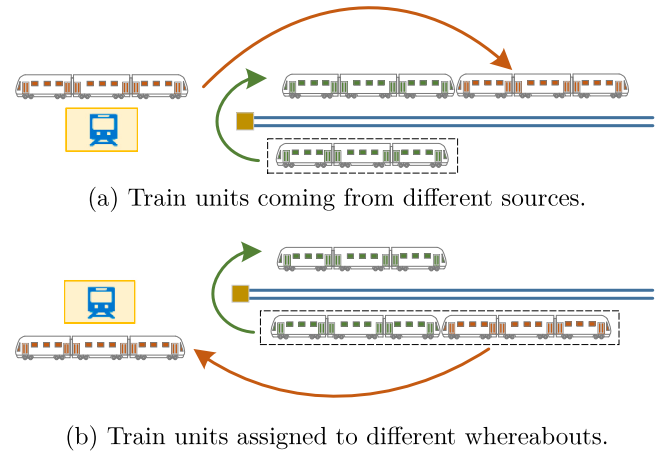


Fig. 3. An illustration of train unit circulation.

compositions of opposite services). As shown in Fig. 3(a), when a train composition starts from a turnaround station, its train units come from an opposite service and the depot. Similarly, when a train composition arrives at a turnaround station, its train units may go to the depot or turn around to form a train composition in the opposite direction (see Fig. 3(b)). Naturally, if a turnaround station is not equipped with a depot, then the train units of a service starting from the station must come from the train compositions of opposite services. Empirically, when a train unit is used to form a new train composition after completing a service, then the minimum turnaround time τ^{turn} must be considered.

2.3. Time-dependent passengers and boarding rules

In this paper, time horizon $[T_s, T_e]$ is discretized into a total of $|T|$ time intervals by a time slot with length Δt , and the set of time stamps is denoted as T (indexed by t). Besides, the passenger demand is pre-given and the set of all passengers is denoted as P (indexed by p). A time-dependent passenger p is described by three attributes, namely *origin station* (o_p), *destination station* (d_p) and the *time stamp entering the URT system* (t_p). According to the locations of origin and destination stations, we define three types of passengers (i.e., *trunk passengers*, *major route passengers* and *minor route passengers*). Wherein, the origin and destination stations of trunk passengers are both on the trunk line, while a major route passenger's origin and destination stations are both on the major route, which are associated with the trunk line and the

major siding line, and a minor route passenger's origin and destination are both on the minor route, which are associated with the trunk line and the minor siding line.

In a few situations, a passenger travels from a station on the major siding line to a station on the minor siding line. In this paper, we pretreat this situation by replacing the passenger with a major route passenger and a minor route passenger. In fact, in addition to the three attributes (o_p, d_p, t_p) of passenger p , we also know the time stamp leaving the URT system of passenger p from the historical data of AFC (Automatic Fare Collection System). Moreover, according to the fixed segment running time, we can calculate the in-vehicle time of each passenger. Then, the historical total waiting time (HWT) of each passenger can be calculated. For a passenger p who travels between the major and minor siding line, we heuristically pretreat him/her as following two consecutive passengers:

- The first passenger's origin station and time stamp entering the URT system are the same as passenger p , while his/her destination station is a joint station.
- The second passenger's destination station is the same as passenger p , while his/her origin station is a joint station, and his/her time stamp entering the URT system is $t_p + 0.5 \cdot \text{HWT} + \chi$, where χ is the first passenger's in-vehicle time.

For the passengers with the same pair of origin and destination stations, their *traveling paths* (i.e., traveling direction and passing-stations) are uniquely determined. We define that train unit route r covers passenger p 's travel path if and only if all his/her passing-stations are on route r . As a result, three conditions should be satisfied if a passenger boards the train unit of a service:

- (1) The passenger's time stamp entering the URT system is not later than the departure time of the service at his/her origin station.
- (2) There is enough spare capacity on the train unit.
- (3) The passenger's travel path is covered by the route of the train unit.

To explain the boarding rules, an example is given in [Appendix A](#).

2.4. Problem definition and assumptions

Given a Y-type URT system and the number of service pairs in the operation time horizon, this paper determines the timetable, train composition of each service and the train unit circulation plan. In this sense, our research problem can be defined as an integrated optimization of TTP and TUCP with consideration of time-dependent passenger demand. To balance service quality and cost, our objective is to minimize passenger waiting time and the number of train units used. To formulate the optimization model, the following assumptions are made:

- **Assumption 1.** The time for coupling/decoupling operation at joint stations, the segment running time and the dwelling time at stations are set as constant values. In general, these values are based on long-term experience to ensure a safe operation. Moreover, the running time between joint stations is not considered because in reality they are the same station.
- **Assumption 2.** All passengers who meet the boarding rules are subject to the FIFO principle.
- **Assumption 3.** Too many train units may be overstocked in a turnaround station to wait to perform the next service, so we assume that the capacity of turnaround stations is unlimited.

3. Model formulation

In this section, we propose a nonlinear integer programming model to optimize the integrated train operation plan on a Y-type URT system with the flexible train composition mode. Some sets used to describe the proposed model are listed in [Table 2](#).

Table 2

Notations for the model.

Notation	Definition
L	Set of lines, $L = \{0, 1, 2\}$, where 0, 1, and 2 represent the trunk line, major siding line and minor siding line respectively.
S_l	Set of stations on line l , $S_l = \{s_{l,1}, s_{l,2}, \dots, s_{l, S_l }\}$, $l \in L$.
S	Set of stations, $S = \bigcup_{l \in L} S_l$.
S^{js}	Set of joint stations, $S^{js} = \{s_{0, S_0 }, s_{1,1}, s_{2,1}\}$.
S^{ts}	Set of turnaround stations, $S^{ts} = \{s_{0,1}, s_{1, S_1 }, s_{2, S_2 }\}$.
$\bar{I}$	Set of downstream services, $\bar{I} = \{1, 2, \dots, \bar{I} \}$, where $ \bar{I} $ is the number of downstream services.
$\bar{I}$	Set of upstream services, $\bar{I} = \{ \bar{I} + 1, \bar{I} + 2, \dots, \bar{I} + \bar{I} \}$, where $ \bar{I} $ is the number of upstream services.
I	Set of services, $I = \bar{I} \cup \bar{I}$.

Table 3

Decision variables.

Variables	Definition
x_i	=1 if service i is a converted service, =0 if service i is a normal service, $\forall i \in I$.
y_i	=1 if service i is performed by a double-composition train on trunk line, = 0 if service i is performed by a single-composition train on trunk line, $\forall i \in I$.
$t_{i,s}^{dep}$	Departure time of service i at station s , $\forall i \in I, s \in S$.
$t_{i,s}^{arr}$	Arrival time of service i at station s , $\forall i \in I, s \in S$.
$c_{i,i',s}$	The number of train units for service i' inherited from service i at station s , $\forall i, i' \in I, s \in S^{ts}$.
$e_{i,s}$	The number of train units for service i entering the depot near station s , $\forall i \in I, s \in S^{ts}$.
$o_{i,s}$	The number of train units for service i obtained from the depot near station s , $\forall i \in I, s \in S^{ts}$.
$z_{p,i,r}$	=1 if passenger p boards the train unit of service i associated with route r , = 0 otherwise, $\forall p \in P, i \in I, r \in R$.

Table 4

Notations for train operations.

Notation	Definition
h^{min}	Minimum headway to ensure safety.
t_s^{dt}	Dwelling time at station s , $s \in S$.
t_s^{rt}	Running time in segment s , $s \in S \setminus \bigcup_{l \in L} \{s_{l, S_l }\}$.
τ^{cd}	Coupling or decoupling time at joint stations.

3.1. Variables

A series of decision variables related to TTP, TUCP and passenger demand are defined and listed in [Table 3](#).

3.2. Constraints

3.2.1. Constraints for train operations

A detailed explanation of the notations used in this section is listed in [Table 4](#).

(1) Headway constraints

In a URT system, headway is defined as the time difference between two consecutive services when arriving at or departing from the same station. In our problem, because the running time between two consecutive stations is assumed to be constant, the arrival headway of two consecutive services at the next station is equal to the departure headway at the previous station. Therefore, we only formulate the departure headway constraints, as follows:

$$t_{i+1,s}^{dep} - t_{i,s}^{dep} \geq h^{min}, \quad \forall i \in I, s \in S. \quad (1)$$

(2) Arrival/departure time constraints

Note that the segment running time and station dwelling time are fixed. If service i 's departure time from station s is given, then it is easy to calculate the arrival and departure times at the following stations.

In this section, we formulate the constraints on downstream services, and that on upstream services can be similarly deduced.

Arrival time: For any downstream service $i \in \bar{I}$ and station $s \in S \setminus \bigcup_{l \in L} \{s_{l,s_l}\}$, the arrival time at next station $s+1$ is formulated as follows,

$$t_{i,s+1}^{arr} = t_{i,s}^{dep} + t_s^{rt}, \quad \forall i \in \bar{I}, s \in S \setminus \bigcup_{l \in L} \{s_{l,s_l}\}. \quad (2)$$

Departure time: For the stations that are not joint stations, the departure time can be calculated by arrival time and station dwelling time, which are formulated as Constraint (3),

$$t_{i,s}^{dep} = t_{i,s}^{arr} + t_s^{dt}, \quad \forall i \in \bar{I}, s \in S \setminus S^{js}. \quad (3)$$

However, because it takes additional time to couple/decouple for the converted services at joint stations, we formulate these constraints for two cases (i.e., normal services and converted services).

Case 1: service i is a normal service,

$$t_{i,s_0,s_0}^{dep} = t_{i,s_0,s_0}^{arr} + t_{s_0,s_0}^{dt}, \quad \forall i \in \bar{I}.$$

Case 2: service i is a converted service,

$$t_{i,s_0,s_0}^{dep} = t_{i,s_0,s_0}^{arr} + t_{s_0,s_0}^{dt} + \tau^{cd}, \quad \forall i \in \bar{I}.$$

Combining Case 1 and Case 2, a comprehensive constraint can be formulated as follows,

$$t_{i,s_0,s_0}^{dep} = t_{i,s_0,s_0}^{arr} + t_{s_0,s_0}^{dt} + \tau^{cd} \cdot x_i, \quad \forall i \in \bar{I}. \quad (4)$$

(3) Joint station constraints

As mentioned earlier, all joint stations are actually the same station in reality, so the departure times at all joint stations of each service are the same,

$$t_{i,s}^{dep} = t_{i,s'}^{dep}, \quad \forall i \in I, s, s' \in S^{js}. \quad (5)$$

3.2.2. Constraints for train compositions

We introduce a set of auxiliary variables $\mu_{i,s,r}$ to represent the number of train units performing service i at station s associated with route r . Note that the minor route ($r = 2$) is always performed by a single-composition if the service is a converted service. This restriction can be formulated as Constraint (6),

$$\mu_{i,s,2} = \eta_{s,2} \cdot x_i, \quad \forall i \in I, s \in S, \quad (6)$$

where $\eta_{s,2}(s \in S)$ is a binary parameter. If station s is on the minor route ($r = 2$), then $\eta_{s,2} = 1$, otherwise $\eta_{s,2} = 0$. The definition of $\eta_{s,1}$ is similar to $\eta_{s,2}$, but on the major route ($r = 1$).

The number of train units performing service i at station s associated with the major route ($r = 1$) equals the total train units of service i less the number of train units associated with the minor route, which is formulated as Constraint (7),

$$\mu_{i,s,1} = \eta_{s,1} \cdot (y_i + 1 - x_i), \quad \forall i \in I, s \in S. \quad (7)$$

Additionally, according to the definition, a converted service must be performed by a double-composition when running on the trunk line, and then Constraint (8) must hold,

$$x_i \leq y_i, \quad \forall i \in I. \quad (8)$$

3.2.3. Constraints for train unit circulation

Table 5 lists the notations used in this section.

(1) Balance constraints of train unit number

At the starting station of a service, the train units of the train composition performing the service may come from a depot or from the train composition that has just turned around. This can be formulated as Constraints (9) and (10) for downstream services and upstream services, respectively,

$$\xi_{s_0,1} \cdot o_{i,s_0,1} + \sum_{i' \in \bar{I}} c_{i',i,s_0,1} = \sum_{r \in R} \mu_{i,s_0,1,r}, \quad \forall i \in \bar{I}, \quad (9)$$

Table 5

Notations for train unit circulation.

Notation	Definition
τ^{turn}	Minimum time for train unit turnarounds.
ξ_s	=1 if there is a depot near turnaround station s , = 0 otherwise, $\forall s \in S^{ts}$.
M	A very large positive number.

Table 6

Notations for passenger demand.

Notation	Definition
P	Set of passengers, $P = \{1, 2, \dots, P \}$, where $ P $ is the number of passengers.
$\bar{P}$	Set of downstream passengers, $\bar{P} \subseteq P$.
$\bar{P}$	Set of upstream passengers, $\bar{P} \subseteq P$.
P_s^{seg}	Set of passengers passing segment s , $s \in S \setminus \bigcup_{l \in L} \{s_{l,s_l}\}$.
P_s^{ori}	Set of passengers with origin station s , $s \in S \setminus \{s_{1,1}, s_{2,1}\}$.
P_t	Set of passengers arriving at their origin station at time t , $t \in T$.
P_r	Set of major and minor route passengers whose travel path is covered by route r , $r \in R$.
o_p	The origin station of passenger p , $p \in P$.
d_p	The destination station of passenger p , $p \in P$.
t_p	The time stamp of passenger p entering the URT system, $p \in P$.
f_p	The first segment traveled by passenger p , $p \in P$.
Cap	Capacity of one train unit.

$$\xi_s \cdot o_{i,s} + \sum_{i' \in \bar{I}} c_{i',i,s} = \sum_{r \in R} \mu_{i,s,r}, \quad \forall i \in \bar{I}, s \in S^{ts} \setminus \{s_{0,1}\}. \quad (10)$$

Similarly, at the last station of a service, the train units of the train composition may go to a depot or turn around to perform services in the opposite direction, which can be formulated as Constraints (11) in the downstream direction and (12) in the upstream direction, respectively,

$$\xi_s \cdot e_{i,s} + \sum_{i' \in \bar{I}} c_{i,i',s} = \sum_{r \in R} \mu_{i,s,r}, \quad \forall i \in \bar{I}, s \in S^{ts} \setminus \{s_{0,1}\}, \quad (11)$$

$$\xi_{s_{0,1}} \cdot e_{i,s_{0,1}} + \sum_{i' \in \bar{I}} c_{i,i',s_{0,1}} = \sum_{r \in R} \mu_{i,s_{0,1},r}, \quad \forall i \in \bar{I}. \quad (12)$$

(2) Conditions of train unit turnarounds

In this section, we introduce auxiliary variable $\rho_{i,i',s}$, where $\rho_{i,i',s} = 1$ represents that the train units of service i' can be inherited from service i at turnaround station s , while $\rho_{i,i',s} = 0$ otherwise.

It must be ensured that the turnaround time is larger than the minimum turnaround time. Therefore, the constraints for minimum turnaround time are formulated as Constraint (13). This represents that if the turnaround time $t_{i',s}^{dep} - t_{i,s}^{arr}$ is less than the minimum turnaround time τ^{turn} , then the train units are not allowed to turnaround (i.e., $\rho_{i,i',s} = 0$),

$$M \cdot (\rho_{i,i',s} - 1) \leq t_{i',s}^{dep} - t_{i,s}^{arr} - \tau^{turn}, \quad \forall i, i' \in I, s \in S^{ts}. \quad (13)$$

Obviously, the number of train units for service i' inherited from service i at station s is greater than zero (i.e., $c_{i,i',s} \geq 1$) only if $\rho_{i,i',s} = 1$, which leads to the following relation:

$$c_{i,i',s} \leq M \cdot \rho_{i,i',s}, \quad \forall i, i' \in I, s \in S^{ts}. \quad (14)$$

3.2.4. Constraints for passenger demand

The notations used in this section are listed in Table 6.

(1) Boarding time constraints

We introduce auxiliary variables $v_{p,i}$. If $v_{p,i} = 1$, the arrival time of passenger p is not later than the departure time of service i at his/her origin station o_p , while $v_{p,i} = 0$ otherwise. According to this definition, Constraint (15) and Constraint (16) are formulated,

$$M \cdot v_{p,i} \geq t_{i,s}^{dep} - t_p, \quad \forall i \in I, s \in S, p \in P_s^{ori}, \quad (15)$$

$$M \cdot (v_{p,i} - 1) \leq t_{i,s}^{dep} - t_p, \quad \forall i \in I, s \in S, p \in P_s^{ori}. \quad (16)$$

If the time condition is not satisfied (i.e., $v_{p,i} = 0$), then passenger p cannot board service i ; that is,

$$\sum_{r \in R} z_{p,i,r} \leq v_{p,i}, \quad \forall p \in P, i \in I. \quad (17)$$

(2) Unique boarding constraints

A passenger can only board one train unit, which can be expressed by Constraint (18),

$$\sum_{i \in I} \sum_{r \in R} z_{p,i,r} = 1, \quad \forall p \in P. \quad (18)$$

For a major route passenger or minor route passenger, the boarding train unit must match his/her travel path. Thus, Constraint (19) is formulated,

$$\sum_{i \in I} z_{p,i,r} = 1, \quad \forall r \in R, p \in P_r. \quad (19)$$

(3) Temporal FIFO

We only list the related constraints for the downstream direction, and that for the upstream direction can be formulated in the same way.

Absolute temporal FIFO of trunk passengers. For two passengers p ($p \in P$) and p' ($p' \in P \setminus \bigcup_{r \in R} P_r$) waiting at station o_p (or $o_{p'}$), if the trunk passenger p' arrives earlier than passenger p , then passenger p' has a higher priority to board the forthcoming service. That is, if passenger p boards service i , then passengers p' must board service i or previous services,

$$\sum_{r' \in R} \sum_{i'=1}^i z_{p',i',r'} \geq \sum_{r' \in R} z_{p,i,r'}, \quad \forall i \in \bar{I}, p \in \bar{P},$$

$$p' \in \bigcup_{i' < t_p} (P_{o_p}^{ori} \cap P_{i'} \cap \bar{P} \setminus \bigcup_{r \in R} P_r). \quad (20)$$

Relative temporal FIFO of major and minor route passengers. For two passengers p and p' ($p, p' \in P_r, r \in R$) with the same origin station and target train units, passenger p' who arrives earlier has a higher priority to board the forthcoming service with a train unit matching their travel paths,

$$\sum_{i'=1}^i z_{p',i',r} \geq z_{p,i,r}, \quad \forall i \in \bar{I}, r \in R, p \in \bar{P} \cap P_r,$$

$$p' \in \bigcup_{i' < t_p} (P_{o_p}^{ori} \cap P_{i'} \cap \bar{P} \cap P_r). \quad (21)$$

(4) Spatial FIFO

For passengers at different stations who meet the boarding time condition of the same train unit of a service, the passengers at stations where the train unit passes earlier have higher priority. That is to say, a passenger will board the train unit whose route matches his/her path once the train unit has spare capacity and the boarding time condition is satisfied. Here, we only show the related constraints for the downstream direction, and the upstream direction can be formulated similarly.

Absolute spatial FIFO of trunk passenger. For a trunk passenger, as long as any train unit of the forthcoming service has spare capacity, then his/her boards it because his/her does not care about the route of the train unit. Of course, the premise is that his/her arrival time should be earlier than the departure time of the service at his/her origin station. Constraint (22) represents these rules. For a trunk passenger p and service i , if the boarding time condition for passenger p is satisfied and there is spare capacity in at least one train unit of service i , then passenger p must board service i or previous services,

$$M \cdot \sum_{i'=1}^i \sum_{r \in R} z_{p,i',r} \geq \sum_{r \in R} Cap \cdot \mu_{i,f_p,r} - \sum_{p' \in \bar{P} \cap P_{f_p}^{seg}} \sum_{r \in R} z_{p',i,r} + M \cdot (v_{p,i} - 1),$$

$$\forall i \in \bar{I}, p \in \bar{P} \setminus \bigcup_{r \in R} P_r. \quad (22)$$

Relative spatial FIFO of major and minor route passengers. In contrast to the trunk passengers, the major route passengers and minor route passengers should pay additional attention to the train unit routes to match their travel paths. Thus, a major or minor route passenger can only board the train unit of a service when it matches his/her path and has spare capacity,

$$M \cdot \sum_{i'=1}^i z_{p,i',r} \geq Cap \cdot \mu_{i,f_p,r} - \sum_{p' \in \bar{P} \cap P_{f_p}^{seg}} z_{p',i,r} + M \cdot (v_{p,i} - 1),$$

$$\forall i \in \bar{I}, r \in R, p \in \bar{P} \cap P_r. \quad (23)$$

In the URT system, because the passengers' tickets are not bound with the seats, the passengers usually board the train unit on the principle of FIFO [27]. If these FIFO constraints are not considered, the objective of minimizing passenger waiting time may lead to unrealistic outcomes. For instance, in a situation of oversaturated passenger demand, long-distance passengers arriving early may have to wait longer at stations to facilitate the boarding of more short-distance passengers, ultimately reducing the overall waiting time of the system.

(5) Capacity constraints

The capacity of each train unit is limited, which means that the number of passengers in each train unit in each segment cannot exceed its maximum quantity,

$$\sum_{p \in P_s^{seg}} z_{p,i,r} \leq Cap \cdot \mu_{i,s,r}, \quad \forall i \in I, r \in R, s \in S \setminus \bigcup_{l \in L} \{s_{l,i}, s_{l,i}\}. \quad (24)$$

3.2.5. Additional operational constraints

Because two directions are considered in this paper, some additional constraints are formulated to make the proposed model more realistic.

(1) Turnaround directions

The turnaround direction of train units at each turnaround station is fixed. For example, at the turnaround station on the trunk line, a downstream service can inherit train units from other services, while no other service can inherit a train unit from it. Conversely, an upstream service cannot inherit train units from other services, but can only provide train units to other services. For other turnaround stations on siding lines, similar logic also exists. Thus, some variables of $c_{i,i',s}$ that do not conform to the actual turnaround direction must be equal to zero, which is represented as Constraint (25),

$$\sum_{i' \in \bar{I}} \sum_{i \in I} c_{i',i,s_{0,1}} + \sum_{i \in I} \sum_{i' \in \bar{I}} \sum_{l \in L \setminus \{0\}} c_{i,i',s_{l,i},s_{l,i}} + \sum_{i \in I} \sum_{i' \in \bar{I}} c_{i,i',s_{0,1}}$$

$$+ \sum_{i' \in \bar{I}} \sum_{i \in I} \sum_{l \in L \setminus \{0\}} c_{i',i,s_{l,i},s_{l,i}} = 0. \quad (25)$$

(2) Passenger directions

Downstream passengers cannot board the train unit of upstream services, and vice versa,

$$\sum_{p \in \bar{P}} \sum_{i \in \bar{I}} \sum_{r \in R} z_{p,i,r} + \sum_{p' \in \bar{P}} \sum_{i' \in \bar{I}} \sum_{r \in R} z_{p',i',r} = 0. \quad (26)$$

3.3. Objective functions

Recall that the objective is to minimize the passengers' total waiting time and the number of train units used, which are formulated in Expressions (27) and (28), respectively. Note that when we set the objective function to minimize the sum of train units used, all train units are expected to perform as many services as possible. We assume that the capacity of turnaround stations is unlimited, thus all train units do not have to be sent back to the depots until no more service needs to be performed. As a result, the objective function to minimize the sum of train units used is equal to minimizing the sum of train units

leaving the depots (or the sum of train units entering the depots) in our problem.

$$\min \sum_{p \in P} \sum_{i \in I} \sum_{r \in R} z_{p,i,r} \cdot (t_{i,o_p}^{dep} - t_p), \quad (27)$$

$$\min \sum_{i \in I} \sum_{s \in S^{ts}} o_{i,s}. \quad (28)$$

To deal with the two objective functions in our model, we use a linear weighted method; that is,

$$\min \alpha \cdot \sum_{p \in P} \sum_{i \in I} \sum_{r \in R} z_{p,i,r} \cdot (t_{i,o_p}^{dep} - t_p) + \beta \cdot \sum_{i \in I} \sum_{s \in S^{ts}} o_{i,s}, \quad (29)$$

where α and β are the pre-specified weights of the two objective functions.

4. Solution method

It should be noted in advance that our algorithms are designed based on a special Y-type URT system, where there is no depot equipped at the stations on the minor siding line. This problem is more complex than a Y-type system with all turnaround stations equipped with a depot because the feasibility of the timetable also needs to be considered. Although our algorithm is designed for a special case, it is easy to extend our method to solve similar problems.

In real-world instances, there are many stations, services and passengers in a Y-type URT system, which makes the proposed non-linear optimization model in Section 3 difficult to solve. Although some linearization techniques can be used to transform the model to a linear one, a lot of auxiliary variables are added that further reduce the solution efficiency of general-purpose ILP solvers.

According to the boarding rules of passengers, we know that when the train timetable and train unit circulation plan are given, the boarding decisions of passengers will be determined accordingly. With this in mind, we decompose the proposed integrated model into two sub-models; that is, TCP (timetable and train unit circulation plan) and PA (passenger assignment). The aim is to solve TCP heuristically and then calculate the total waiting time by PA.

The adaptive large neighborhood search (ALNS) algorithm is a heuristic proposed by Pisinger and Ropke [53], which has been applied to many classical large-scale discrete optimization problems successfully, such as vehicle routing problems [53–56], hub location problems [57,58], and packing problem [59,60]. Furthermore, it has been proven that the ALNS algorithm can effectively solve the train timetable problem [61,62]. In this section, we will introduce how to solve the proposed model through the ALNS algorithm, which is improved on the classical framework. We divide this section into two parts. The first part will introduce the solution framework, and the second part will introduce the algorithm designs for our problem.

4.1. Solution framework

In the ALNS algorithm, there are two solutions recorded (i.e., the best-known solution and the current solution). The best-known solution is the best solution until the computation time limit, and the current solution is used as the input of the next iteration, which is set to escape from local optimal since it can be updated to a low-quality solution by a probability. In the classical framework of ALNS, some consecutive identical iterations can be addressed by the asynchronous computation mechanism to improve efficiency. Thus, we propose an improved framework (see Fig. 4), in which we divided it into a global level and an asynchronous level. At the asynchronous level, we explore multiple new solutions rather than one, which means that the framework of one-iteration-one-evaluation is changed to multiple-iterations-one-evaluation. In addition to improving efficiency through asynchronous computing, generating multiple solutions between two evaluations can avoid a premature escape from the current solution.

Our algorithm has several key parameters. As in the classical framework of ALNS, we set *score* (θ^d, θ^r) and *weight* (ϑ^d, ϑ^r) of operators. In addition, we introduce the simulated annealing mechanism to select a low-quality solution for the current solution to avoid local optimum, so two more parameters are used; that is, *temperature* (denoted as *tem*) and *probability* (denoted as *prob*). When the best-known solution or the current solution gets updated, *tem* is initialized to a constant value. Meanwhile, *tem* is updated by reducing Δtem in each iteration. Moreover, the current solution is only allowed to select a low-quality solution according to parameter *prob* when *tem* is non-positive.

In Fig. 4, the algorithm details of the processes in yellow boxes in the asynchronous level and the global level are illustrated in Algorithm 1 and Algorithm 2, respectively. Algorithm 1 shows how to obtain the solution pool through asynchronous multi-threading computing. First, we define the number of threads and then we assign an iteration task to each thread. In each iteration, the current solution $\bar{x}$ is destroyed by a randomly selected destroy operator, the destroyed solution is then repaired by a randomly selected repair operator, and finally, a new solution is obtained. When all threads complete the iterations, the new solutions generated by all threads form a solution pool, which is returned to the global level. In this process, F^d and F^r are a series of destroy operators and repair operators, and V is to calculate the objective function value of the generated solution.

Algorithm 2 is used to evaluate the solutions and update parameters. By comparing each solution in the solution pool with the best-known solution and the current solution, the related values get updated, including the best-known solution, the current solution, and the parameters of the operators. The notations G^d and G^r represent the rules to update the parameters of destroy operators and repair operators, respectively.

Algorithm 1: The steps of iterations for asynchronous multi-threading.

Data: The current solution ($\bar{x}$), weights (ϑ^r, ϑ^d) of repair and destroy operators, number of threads (N), and global iterations counter (k).

Result: A solution pool ($X = \bigcup_{n=1}^N \{(x_n^k, v_n^k, d_n^k, r_n^k)\}$).

Initialize solution pool, $X \leftarrow \emptyset$;

$n \leftarrow 1$;

while $n \leq N$ **do**

 Start thread n ;

 Select a destroy operator (d_n^k) according to ϑ^d ;

$x_n^k \leftarrow F_{d_n^k}^d(\bar{x});$ /* Execute the selected destroy operator. */

 Select a repair operator (r_n^k) according to ϑ^r ;

$x_n^k \leftarrow F_{r_n^k}^r(x_n^k);$ /* Execute the selected repair operator. */

$v_n^k \leftarrow V(x_n^k);$ /* Calculate the objective function value. */

$X \leftarrow X \cup \{(x_n^k, v_n^k, d_n^k, r_n^k)\};$

 End thread n ;

$n = n + 1$.

end

4.2. Algorithm designs

On the basis of our ALNS framework, we introduce the details of our algorithm in this section.

4.2.1. Feasibility proposition

As stated earlier, not all timetables are feasible to generate a train unit circulation plan. To obtain a feasible train unit circulation plan, some propositions must be satisfied.

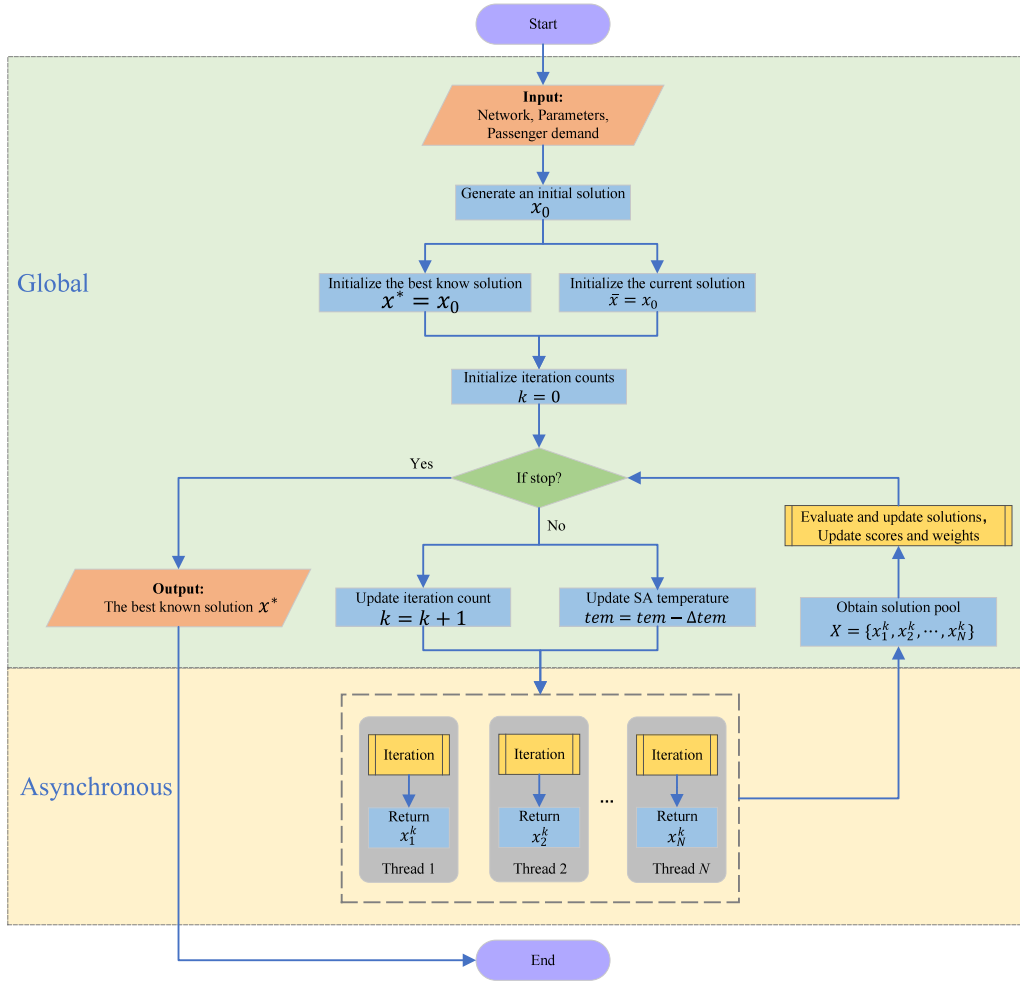


Fig. 4. The framework of our ALNS. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Proposition 1. Only the services starting during the downstream (upstream) minor route time horizon $[T_s, T_e - 2\tau^{tr} - \tau^{turn}]$ ($[T_s + \tau^{tr} + \tau^{turn}, T_e - \tau^{tr}]$) are allowed to be converted services.

Because there is no depot on the minor siding line to provide train units, the first service starting from station $s_{2,|S_2|}$ can only wait until one train composition of an opposite service arrives at station $s_{2,|S_2|}$. Similarly, the time of the last service arriving at station $s_{2,|S_2|}$ should be earlier than that of the last service starting from the station because the train units must return to the depot at station $s_{1,1}$. Assuming that the travel time of a converted service is τ^{tr} and the minimum turnaround time is τ^{turn} , the downstream (upstream) minor route time horizon is $[T_s, T_e - 2\tau^{tr} - \tau^{turn}]$ ($[T_s + \tau^{tr} + \tau^{turn}, T_e - \tau^{tr}]$).

Proposition 2. Minimum number of train units on line l must satisfy

$$N_l = \text{CEIL} \left\{ \text{MAX} \left\{ \frac{\text{COUNT}(P_s^{seg})}{Cap}, s \in S_l \setminus \{s_{l,|S_l|}\} \right\} \right\}, \quad \forall l \in L. \quad (30)$$

To ensure that all passengers are served, the minimum number of train units on each line needs to be considered in advance. In essence, the segment with the largest passenger demand determines the minimum number of train units required on that line. Eq. (30) is used to calculate the minimum number of train units for line l , where COUNT is a function that returns the length of a set and function CEIL returns the smallest integer that is not smaller than the given variable.

Proposition 3. The last service starting during the downstream minor route time horizon $\bar{i}^l$ must satisfy

$$\text{MAX} \left\{ \bar{N}_2, |\bar{I}| - \frac{\tau^{tr} + \tau^{turn}}{h^{min}} \right\} \leq \bar{i}^l \leq \frac{T_e - 2\tau^{tr} - \tau^{turn} - T_s}{h^{min}} + 1. \quad (31)$$

According to Proposition 2, the minimum number of train units on the minor siding line in the downstream direction is $\bar{N}_2$, so there are at least $\bar{N}_2 - 1$ converted services before service $\bar{i}^l$, and then Eq. (32) holds,

$$\bar{i}^l \geq \bar{N}_2. \quad (32)$$

Moreover, Eq. (33) represents the headways of downstream services starting during the downstream minor route time horizon cannot be less than the minimum headways h^{min} ,

$$(\bar{i}^l - 1) \cdot h^{min} \leq T_e - 2\tau^{tr} - \tau^{turn} - T_s. \quad (33)$$

Similarly, the headways of downstream services starting outside the downstream minor route time horizon cannot be less than the minimum headways h^{min} , which is formulated as Eq. (34),

$$(|\bar{I}| - \bar{i}^l) \cdot h^{min} \leq \tau^{tr} + \tau^{turn}. \quad (34)$$

By solving Eqs. (32)–(34), Proposition 3 is derived.

Algorithm 2: Steps to evaluate and update solutions, scores and weights at iteration k .

Data: A solution pool ($X = \bigcup_{n=1}^N \{(x_n^k, v_n^k, d_n^k, r_n^k)\}$), the best-known solution (x^*), the current solution ($\bar{x}$), weights (ϑ^d , ϑ^r) of destroy and repair operators, the temperature (tem), and accepted probability ($prob$).

Result: Update the solutions and parameters of ALNS.

Initialize candidate solution set, $\mathbb{S} \leftarrow \emptyset$;
 $n \leftarrow 0$;
while $n \leq N$ **do**
 if x_n^k is better than x^* **then**
 $x^* \leftarrow x_n^k$;
 $\bar{x} \leftarrow x_n^k$;
 Initialize tem ;
 $\theta_{d_n}^d \leftarrow G_1^d(d_n^k)$; /* Measure the score of the
 executed destroy operator. */
 $\theta_{r_n}^r \leftarrow G_1^r(r_n^k)$; /* Measure the score of the
 executed repair operator. */
 $\mathbb{S} \leftarrow \mathbb{S} \cup \{(x_n^k, v_n^k)\}$;
 else if x_n^k is better than $\bar{x}$ **then**
 $\bar{x} \leftarrow x_n^k$;
 Initialize tem ;
 $\theta_{d_n}^d \leftarrow G_2^d(d_n^k)$;
 $\theta_{r_n}^r \leftarrow G_2^r(r_n^k)$;
 $\mathbb{S} \leftarrow \mathbb{S} \cup \{(x_n^k, v_n^k)\}$;
 else if x_n^k is accepted as a candidate solution **then**
 if $tem \leq 0$ **then**
 $\mathbb{S} \leftarrow \mathbb{S} \cup \{(x_n^k, v_n^k)\}$;
 end
 $\theta_{d_n}^d \leftarrow G_3^d(d_n^k)$;
 $\theta_{r_n}^r \leftarrow G_3^r(r_n^k)$;
 else
 $\theta_{d_n}^d \leftarrow G_4^d(d_n^k)$;
 $\theta_{r_n}^r \leftarrow G_4^r(r_n^k)$;
 end
 if $tem \leq 0$ and $\mathbb{S}$ is not $\emptyset$ **then**
 Select a solution in candidate solution set $\mathbb{S}$ as current solution randomly.
 end
 Update ϑ^r and ϑ^d by θ^r and θ^d .

Proposition 4. The first service starting during the upstream minor route time horizon $\bar{t}^f$ must satisfy

$$|\bar{t}| + |\bar{t}| - \frac{T_e - 2\tau^{tr} - \tau^{turn} - T_s}{h^{min}} \leq \bar{t}^f$$

$$\leq \min \left\{ |\bar{t}| + |\bar{t}| + 1 - \bar{N}_2, \frac{\tau^{tr} + \tau^{turn}}{h^{min}} + |\bar{t}| + 1 \right\}. \quad (35)$$

Proposition 4 is derived similarly to Proposition 3.

4.2.2. Initial solution generation

In our problem, an initial solution is a set of information satisfying the constraints for a train timetable and train unit circulation plan, which includes all variables in Table 3 except $z_{p,i,r}$. The following lemma gives a sufficient condition for generating feasible solutions.

Lemma 4.1. If the number of services during the downstream (upstream) minor route time horizon in Section 4.2.1 is $\bar{N}(\bar{N})$, and the headway

between downstream (upstream) services is $\bar{h}(\bar{h})$, where

$$\bar{h} = \frac{T_e - T_s - 2\tau^{tr} - \tau^{turn}}{\bar{N} - 1}, \quad (36)$$

$$\bar{h} = \frac{T_e - T_s - 2\tau^{tr} - \tau^{turn}}{\bar{N} - 1}, \quad (37)$$

then a feasible solution with $\underline{N}(\underline{N} = \min\{\bar{N}, \bar{N}\})$ pairs of converted services can always be generated.

Proof. Note that the dwelling time at turnaround stations is set as zero by default, so the departure and arrival times of each service at turnaround stations are equivalent. We discuss them in the two following possible cases.

- If $\bar{N} \geq \bar{N}$, then $\underline{N} = \min\{\bar{N}, \bar{N}\} = \bar{N}$. According to Eqs. (36) and (37), $\bar{h} \leq \bar{h}$ holds. Denote by $\bar{t}_i^+$ ($\bar{t}_i^-$) the arrival (departure) time of the i th downstream (upstream) service at station $s_{2,|S_2|}$, we then calculate the departure/arrival time of services at station $s_{2,|S_2|}$,

$$\bar{t}_i^+ = T_s + \tau^{tr} + (i - 1) \cdot \bar{h}, \quad \forall i = 1, 2, \dots, \underline{N},$$

$$\bar{t}_i^- = T_s + \tau^{tr} + \tau^{turn} + (i - 1) \cdot \bar{h}, \quad \forall i = 1, 2, \dots, \underline{N}.$$

By subtracting the former from the latter of the above two equations, we have

$$\bar{t}_i^- - \bar{t}_i^+ = \tau^{turn} + (i - 1) \cdot (\bar{h} - \bar{h}) \geq \tau^{turn}, \quad \forall i = 1, 2, \dots, \underline{N}.$$

- If $\bar{N} < \bar{N}$, then $\underline{N} = \min\{\bar{N}, \bar{N}\} = \bar{N}$. According to Eqs. (36) and (37), $\bar{h} > \bar{h}$ holds. Denote by $\bar{t}_i^-$ ($\bar{t}_i^+$) the arrival (departure) time of the reverse i th downstream (upstream) service at station $s_{2,|S_2|}$, we then calculate the departure/arrival time of services at station $s_{2,|S_2|}$,

$$\bar{t}_i^- = T_e - \tau^{tr} - \tau^{turn} - (i - 1) \cdot \bar{h}, \quad \forall i = 1, 2, \dots, \underline{N},$$

$$\bar{t}_i^+ = T_e - \tau^{tr} - (i - 1) \cdot \bar{h}, \quad \forall i = 1, 2, \dots, \underline{N}.$$

By subtracting the former from the latter of the above two equations, we have

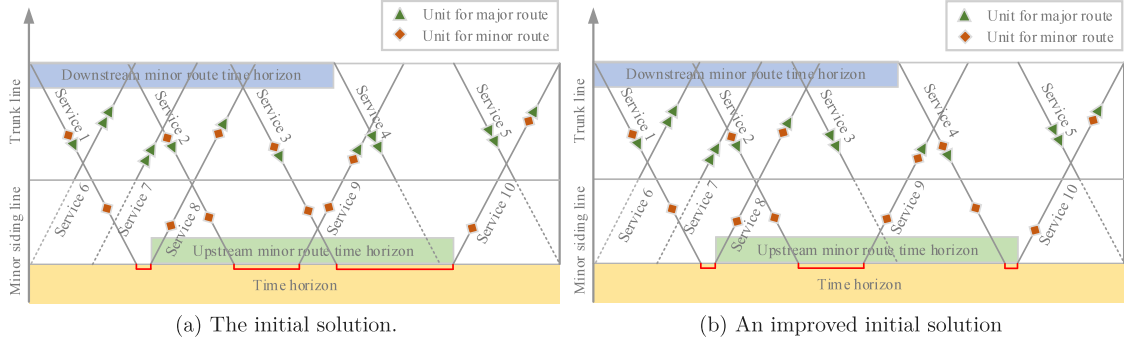
$$\bar{t}_i^- - \bar{t}_i^+ = \tau^{turn} + (i - 1) \cdot (\bar{h} - \bar{h}) \geq \tau^{turn}, \quad \forall i = 1, 2, \dots, \underline{N}.$$

We conclude that a feasible solution always holds by selecting $\underline{N}$ services as converted services in each direction in order ($\bar{N} \geq \bar{N}$) or in reverse order ($\bar{N} < \bar{N}$). This lemma is thus proven. $\square$

The method to randomly generate an initial solution can be stated as follows,

- (1) Randomly select $\bar{t}^l$ and $\bar{t}^f$ satisfying Propositions 3 and 4.
- (2) Services starting during the downstream (upstream) minor route time horizon are scheduled with constant headway $\bar{h}(\bar{h})$.
- (3) Set all services to be performed by a double-composition when running on the trunk line.
- (4) Select the services to be converted services according to Lemma 4.1.

In Fig. 5(a), an example with 10 services is used to illustrate these steps when $\bar{N} \geq \bar{N}$. Service 4 is selected as the last service starting during the downstream minor route time horizon (i.e., $\bar{t}^l = 4$), while service 8 is selected as the first service starting during the upstream minor route time horizon (i.e., $\bar{t}^f = 8$). The services starting during/outside the downstream/upstream minor route time horizon are then scheduled with average headway. Furthermore, all services are set to be performed by a double-composition when running on the trunk line, which is represented as two triangles or a triangle and a diamond. Finally, because $\bar{N} \geq \bar{N}$ holds, three pairs of services are selected as converted services in order (i.e., services 1, 2, and 3, and services 8, 9, and 10).

Fig. 5. Generating an initial solution ($\bar{N} \geq \bar{N}$).

Although the solution represented by Fig. 5(a) is feasible for a train timetable and train unit circulation plan, it may mean that some passengers cannot be served because the minor siding line in downstream direction ends the operation prematurely. To avoid such potential defects and make it more in line with the actual operation, we set a precondition that both the first service and last service during the downstream/upstream minor route time horizon are converted services. We then select the other $\bar{N} - 2$ pairs of services as converted services according to Lemma 4.1. As shown in Fig. 5(b), services 1, 4, 8, and 10 are first selected as converted services, followed by service 2 and service 9.

4.2.3. Operator design

The destroy operators and repair operators are used to search for another solution around the current solution, which are the key elements of the framework of ALNS. In our problem, the core task is to search for a reasonable timetable and train unit circulation plan. Thus, our operators are mainly designed based on these two points.

(1) Destroy operators

Destroy 1: Randomly change $\bar{i}^l$ and $\bar{i}^f$.

This operator is equivalent to generating the initial solution, and Section 4.2.2 gives the steps.

Destroy 2: Randomly change the number of converted services.

In this operator, we first randomly select a service from all services. If the selected service is a converted service, then we find another service that generates the train unit turnaround with the selected service at station $s_{2,|S_2|}$ in the current solution. We then change the two converted services into normal services. If the selected service is a normal service, we find another opposite normal service that is available to generate train unit turnaround with the selected service at station $s_{2,|S_2|}$ in time dimension, and then both of them are changed into converted services. Obviously, only if such two normal services are found can we change them into converted services. This operator actually changes the number of converted services (as well as the normal services), which contains two sub-operators. One is reducing-operator to change one pair of converted services to normal services, the other is adding-operator to change one pair of normal services to converted services. Whether these two sub-operators are executed in each iteration is determined by a preset probability. Note that if a normal service performed by a single-composition is selected to implement adding-operator, its train composition should change to a double-composition.

An illustration of Destroy 2 is shown in Fig. 6, where Fig. 6(a) gives the current solution. In an iteration, we assume that service 2 is selected to be applied by Destroy 2. Due to the fact that service 2 is a converted service, the reducing-operator is used to change service 2 and service 9 to normal services, then a solution is generated (see Fig. 6(b)). Furthermore, if the adding-operator is executed on Fig. 6(b), then another solution is generated, which is one of the three outcomes

of Figs. 6(a), (c), and (d). We can see that the outcome of Fig. 6(d) changes the train composition of service 3.

Destroy 3: Randomly change the train compositions of normal services.

Because there are depots at both ends of the major route, changing the train composition of normal services will not affect the feasibility. This operator is used to randomly select a normal service i (where $x_i = 0$ in the current solution) and let $y_i = 1 - y_i$. For example, if the current solution is shown as Fig. 6(a), then service 3, 5, 6 and 8 can be selected by this operator.

Destroy 4: Randomly shift the arrival/departure times of a service.

This operator randomly selects a service and shifts its arrival/departure times. Note that the headway cannot be less than the minimum headway h^{min} , and the train unit turnarounds in the current solution cannot be destroyed to avoid infeasible solutions. Therefore, the shifting time range is limited not only by the headway but also by the train unit turnarounds at station $s_{2,|S_2|}$.

As an example, consider the current solution in Fig. 7(a). Assume that service 3 is selected, then its allowed shifting time range is shown as the pink parallelogram box in Fig. 7(b). The left bound is limited by the headway constraint of service 2 and service 3 as well as the train unit turnaround constraint of service 6 and service 3, while the right bound is limited by the headway constraint of service 3 and service 4 as well as the train unit turnaround constraint of service 3 and service 9.

(2) Repair operators

Repair 1: Reduce the maximum train unit turnaround time on the minor siding line.

It can be inferred that after many iterations, the train unit of a downstream converted service may have to wait a long time to perform the next service at station $s_{2,|S_2|}$. Generally, this scenario will make the turnaround time of the train unit too long, which results in the use of more train units. Moreover, this makes it difficult to find a pair of normal services to change into converted services. Therefore, a repair operator is set to improve it, aiming to reduce the maximum turnaround time.

This operator is executed by changing the train unit turnarounds. In this operator, we first find the service pair with maximum train unit turnaround time at station $s_{2,|S_2|}$. We then change one of them into a normal service. Finally, change another normal service into a converted service, which is available to generate a train unit turnaround with a shorter turnaround time for the rest of the converted service. Recall the examples in Fig. 6, this operator can convert Fig. 6(a) into Fig. 6(c) or Fig. 6(d) directly.

Repair 2: Add converted services.

Although we consider the minimum number of converted services when generating the initial solution, Destroy 2 may reduce the converted services and make it impossible to serve all passengers. Moreover, fewer converted services result in a larger waiting time of minor route passengers. To deal with this condition, which may make the

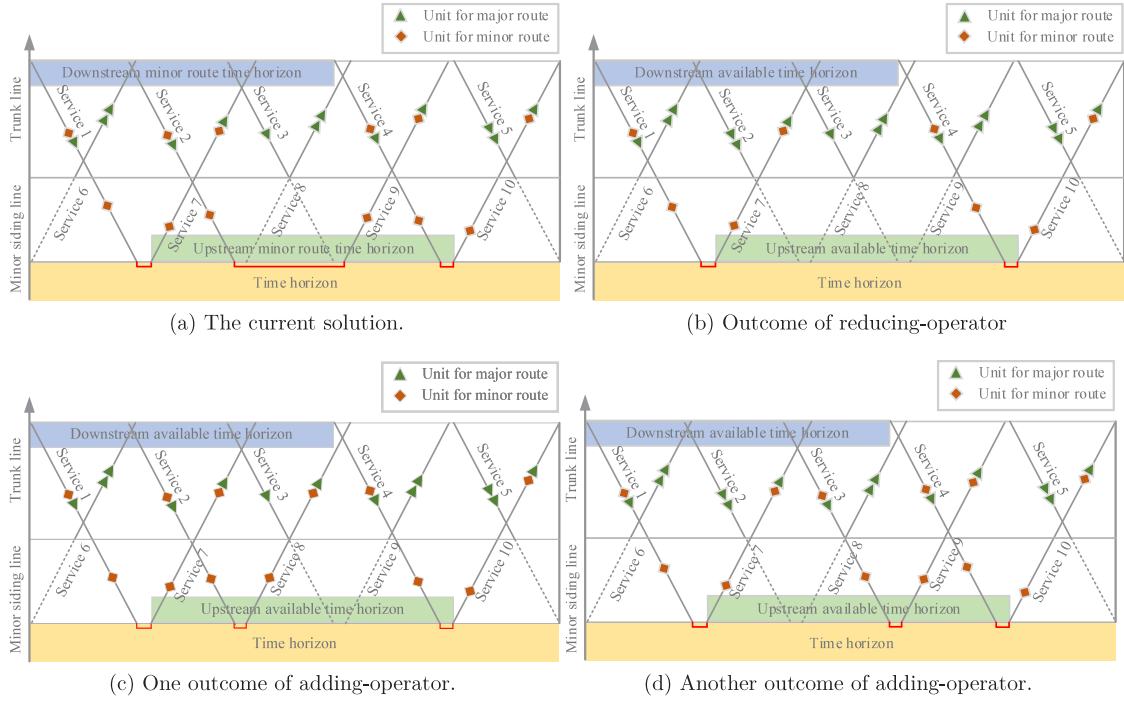


Fig. 6. An illustration of Destroy 2.

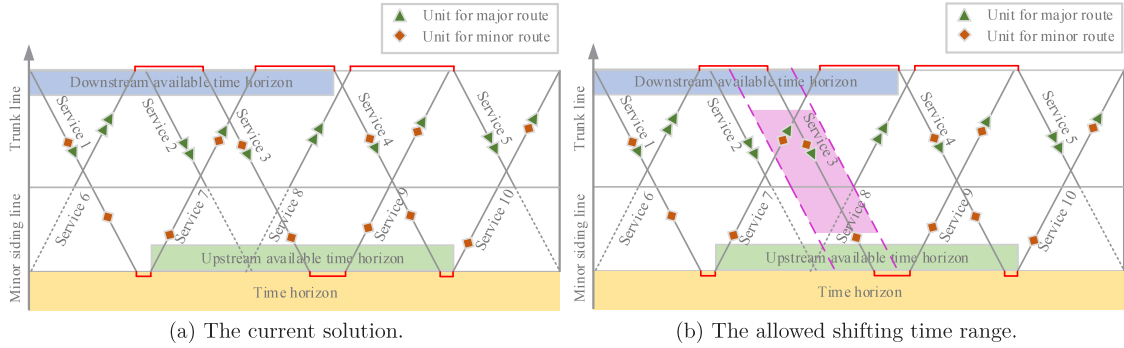


Fig. 7. An illustration of Destroy 4.

objective function worse, we change one pair of normal services into converted services in this step, which is the same as **Destroy 2** under the condition that the selected service is a normal service.

Repair 3: Reduce the maximum headway of services.

A headway that is too large between two consecutive services will affect the waiting time of passengers and the train unit turnarounds. To reduce the headway, this operator makes two consecutive services with the maximum headway shift toward each other. As **Destroy 4**, this operator also cannot destroy the train unit turnarounds to avoid generating infeasible solutions. Thus, the allowed shifting time range of the two services should be calculated in advance by the headway, as well as the train unit turnarounds.

4.2.4. Passenger assignment (PA)

We search for the neighboring solutions of TCP heuristically through different operator combinations. In our problem, if the train timetable is given, then the passenger assignment is also determined due to the route-based FIFO boarding rules stated in Section 2.3. Thus, a simulation-based method is adopted to assign the passengers to services, which is based on Wu et al. [27]. We only show the assignment

procedure for downstream passengers in Algorithm 3, and that of upstream passengers is similar. Several sets are introduced, which can easily be obtained through the passenger demand and the heuristic solution of TCP. Set $U = \{u_{i,s} | u_{i,s} = Cap \cdot \sum_{r \in R} \mu_{i,s,r}, i \in I, s \in S\}$, where $u_{i,s}$ represents the initial capacity provided by service i at station s , $\bar{P}_{t,s} (\forall t \in T, s \in S)$ represents the set of downstream passengers whose origin station is station s and arrival time is time stamp t , and $\Lambda_p (\forall p \in P)$ is the set of segments that passenger p passes through.

4.2.5. Adaptive searching strategy

We adopt a standard adaptive searching strategy in ALNS [63] to select the most effective operators. The scores (θ^d, θ^r) are set to measure the potential contributions to iterations. The weights $(\vartheta^d, \vartheta^r)$ reflect the importance of each operator in the whole process, which are updated by the scores in each iteration. When initializing algorithm parameters, all weights are set to 1 and all scores are set to 0.

(1) Measure scores

In each iteration, some operators are selected randomly according to the weights. The scores of the selected operators are then measured based on the quality of the generated solution. Taking the k th destroy

Algorithm 3: Improved procedure for downstream passenger assignment.

Data: Departure time $t_{i,s}^{dep} (\forall i \in \bar{T}, s \in S)$, set $T, \bar{T}, U$,

$\bar{P}_{i,s} (\forall i \in T, s \in S), \Lambda_p (\forall p \in P)$.

Result: The result of downstream passenger assignments.

Initialize the sets, $T_s^{min} (\forall s \in S) \leftarrow T$;

```

for  $i$  in  $\bar{T}$ ,  $s$  in  $S$  do
  for  $t = \text{MIN}(T_s^{min})$  to  $t_{i,s}^{dep}$  do
    for  $p$  in  $\bar{P}_{i,s}$  do
      Initialize a boarding flag,  $flag \leftarrow 1$ ;
      for  $seg$  in  $\Lambda_p$  do
        if  $\mu_{i,seg} \leq 0$  then
           $flag \leftarrow 0$ ;
          Break;
        end
      end
      if  $flag == 1$  then
        for  $seg$  in  $\Lambda_p$  do
           $u_{i,seg} \leftarrow u_{i,seg} - 1$ ;
        end
        Assign passenger  $p$  to train  $i$ ;
        Remove  $p$  from set  $\bar{P}_{i,s}$ ;
      end
    end
    if  $\bar{P}_{i,s}$  is  $\emptyset$  then
      Remove  $t$  from set  $T_s^{min}$ ;
    end
  end
end

```

Output the assignment results of downstream passengers.

operator as an example, its score in one iteration is shown as follows,

$$\theta_k^d = \begin{cases} score_1, & \text{if the generated solution is better than} \\ & \text{the best solution;} \\ score_2, & \text{if the generated solution is better than} \\ & \text{the current solution;} \\ score_3, & \text{if the generated solution is accepted;} \\ score_4, & \text{if the generated solution is rejected,} \end{cases}$$

where parameters $score_1, score_2, score_3$ and $score_4$ meet $score_1 \geq score_2 \geq score_3 \geq score_4$.

(2) Update weights

After the scores of operators are measured, the weights of operators are updated by the scores. The update method is based on the following formulas,

$$\theta_k^d = \lambda \theta_k^d + (1 - \lambda) \theta_k^d,$$

$$\theta_k^r = \lambda \theta_k^r + (1 - \lambda) \theta_k^r.$$

In our problem, two roulette wheel mechanisms are used to adaptively select operators through the weights of the operators. One is used to select a destroy operator, and the other is used to select a repair operator. Take a destroy operator as an example, the probability of selecting Destroy i is calculated by

$$\theta_i^d / \sum_{k=1}^{|O_d|} \theta_k^d,$$

where $|O_d|$ is the number of destroy operators. Similarly, if the number of repair operators is $|O_r|$, then the probability of selecting Repair i is calculated by

$$\theta_i^r / \sum_{k=1}^{|O_r|} \theta_k^r.$$

4.2.6. Stopping criteria

In such a heuristic algorithm, it is difficult to guarantee the optimality of the obtained solutions. Therefore, whether the algorithm needs to be stopped cannot be determined by the quality of the solutions. Here, we introduce the stopping criteria in [61], two key items are as follows,

- (1) The iteration count, or the computation time reaches the preset maximum value.
- (2) The time to search for the next better solution reaches the preset maximum value.

When one of the two items has been met, the algorithm stops. Item (1) limits the maximum iteration count and time of the algorithm, which is the most basic stopping condition and is widely used in heuristic algorithms. Item (2) limits the time to search for a better solution. This happens because the best-known solution is so good that it is difficult to search for a better solution in a given long time.

5. Computational experiments

In this section, we choose Guangzhou Metro Line 14 (see Fig. 8) as our real-world case study to verify the effectiveness of our algorithm. Considering the influence of the algorithm parameters on experimental efficiency and accuracy, we first generate a group of experiments to test the performance of different parameters. Then, to show the applicability of our method in real-world cases, we design a group of experiments to test the performance of our method under different passenger demand scenarios. Additionally, we compare the objective function values under the traditional fixed train composition mode and the flexible train composition mode. Finally, we compare the performance of the proposed ALNS algorithm and a general-purpose ILP solver by constructing a group of small-case instances. It should be noted that we linearize the nonlinear model formulated in Section 3 to a linear model, and the detailed process is shown in Appendix B. Experiments are all coded by Python on a personal computer with an AMD Ryzen 5 3600X 6-core 12-thread Processor @ 3.80 GHz processor, 16 GB RAM and Windows 10 operating system. GUROBI 9.1.0 is used as the ILP solver to solve the models in small-case instances.

5.1. Description of real-world case and parameter settings

Guangzhou Metro Line 14 is selected as our real-world case study, and the time horizon is set to 4 hours with a time slot of 30 seconds. During the time horizon, 80 services (40 pairs) will be scheduled. The relevant parameter settings are listed in Tables C.1 and C.2 in Appendix C. Three passenger demand scenarios are given, named as POP, UAT, and MHMT, each of which loads 97 008 passengers with variant temporal and/or spatial distribution. In the POP scenario, the time horizon is divided into peak-period and off-peak period according to the density of passenger demand. In the UAT scenario, passengers arrive at origin stations with uniform density in the temporal dimension. In the MHMT scenario, designed to evaluate the performance of our algorithm in an extreme passenger demand context, we halve the passenger demand density on the minor siding line and double it on the major siding line, relative to the POP scenario. The MHMT scenario is founded on the premise that areas along the major siding line are accorded greater importance in future residential land development plans than areas along the minor siding line. Given this context, the operating department of a Y-type URT system is inclined to enhance passenger demand on the major siding line while scaling down passenger demand on the minor siding line during the passenger demand forecasting process. The passenger demand of each scenario is presented in Appendix D. In reality, the objective function weights can be set to different values based on the user's preference, wherein the larger the weight, the higher the preference. We have tested a group of experiments by changing the values of α and β . The results show that it is more in line with the actual operational requirements to take the values of 1 and 10 000, respectively. Besides, α and β can keep two objective functions on the same order of magnitude, in case one of them dominates.

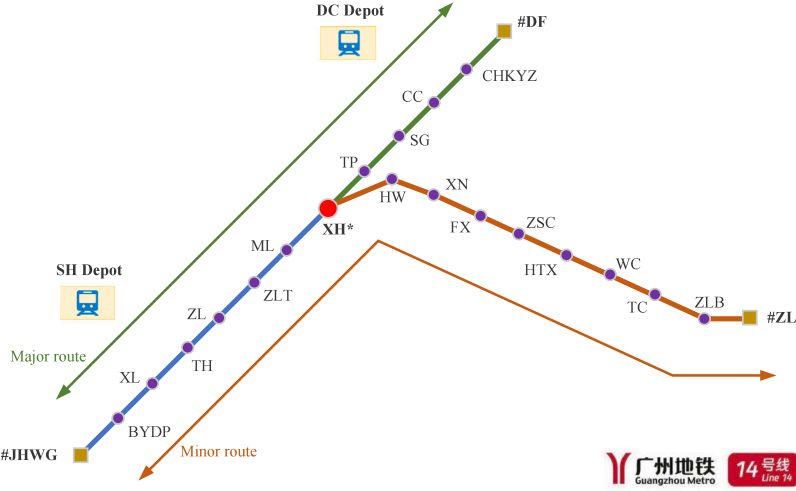


Fig. 8. Guangzhou Metro Line 14.

Table 7
Computational results with different parameters.

Instance	OBJ	TWT ^b [min]	# TU	CPU [s]	Gap
<i>PAP</i> – 1 – 0.5	888 419	214 209	46	2686.40	3.31%
<i>PAP</i> – 1 – 0.05	879 179	239 589	40	1211.62	2.24%
<i>PAP</i> – 1 – 0.005	879 036	224 518	43	1894.42	2.22%
<i>PAP</i> – 1 – 0.0005	877 631	223 815	43	3204.62	2.06%
<i>PAP</i> – 1 – 0	883 382	221 691	44	3358.76	2.73%
<i>PAP</i> – 3 – 0.5	885 515	232 757	42	1238.30	2.97%
<i>PAP</i> – 3 – 0.05	878 608	209 304	46	1414.75	2.17%
<i>PAP</i> – 3 – 0.005	859 942 ^a	229 971	40	1266.29	0
<i>PAP</i> – 3 – 0.0005	888 317	214 158	46	2267.43	3.30%
<i>PAP</i> – 3 – 0	878 095	214 047	44	3249.76	2.11%
<i>PAP</i> – 5 – 0.5	886 509	258 254	37	2745.78	3.09%
<i>PAP</i> – 5 – 0.05	872 395	241 197	39	1952.55	1.45%
<i>PAP</i> – 5 – 0.005	875 172	217 586	44	349.32	1.77%
<i>PAP</i> – 5 – 0.0005	864 806	212 403	44	1332.65	0.57%
<i>PAP</i> – 5 – 0	907 191	208 595	49	3174.62	5.49%
<i>PAP</i> – 7 – 0.5	864 806	212 403	44	2719.05	0.57%
<i>PAP</i> – 7 – 0.05	886 509	258 254	37	658.11	3.09%
<i>PAP</i> – 7 – 0.005	864 806	212 403	44	394.38	0.57%
<i>PAP</i> – 7 – 0.0005	883 373	231 686	42	2762.88	2.72%
<i>PAP</i> – 7 – 0	865 191	212 595	44	2710.47	0.61%
<i>PAP</i> – 9 – 0.5	873 915	216 957	44	3123.97	1.62%
<i>PAP</i> – 9 – 0.05	864 724	217 362	43	1336.23	0.56%
<i>PAP</i> – 9 – 0.005	877 917	218 958	44	2232.66	2.09%
<i>PAP</i> – 9 – 0.0005	864 806	212 403	44	2810.28	0.57%
<i>PAP</i> – 9 – 0	930 848	220 424	49	3342.54	8.25%

^a The best-obtained objective function value in these instances.

^b Note that a time slot is 30 s in these instances, while the unit of TWT is minute, thus $TWT = \frac{OBJ - \beta \cdot TU}{2\alpha}$.

5.2. Performance of the ALNS algorithm with different parameters

In the POP scenario, we design a total of 25 instances by setting different algorithm parameters, which are distinguished by the various combinations of parameters *tem* and *prob*. These instances are referred to as *PAP* – *tem* – *prob* (e.g., *PAP* – 1 – 0.5), corresponding to the instance with *tem* = 1 and *prob* = 0.5.

The experimental results are reported in Table 7, including the objective function value (OBJ), total waiting time of passengers (TWT), the number of train units used (# TU), and the computation time to find the best-known solution (CPU). Because the ALNS algorithm is an adaptive heuristic, which does not guarantee the optimality of the obtained solutions, we only analyze the best-known solutions in these instances. Among all these results, the best-obtained solution is from instance *PAP* – 3 – 0.005, with an objective function value of 859 942.

To compare the experimental results more intuitively, we calculate the gap between the objective function value of each instance and the best-obtained objective function value, which is shown in the last column of Table 7. It can be seen that the gaps of most solutions are small enough (less than 3%), which indicates the stability of our algorithm.

Note that the instances *PAP* – 5 – 0 and *PAP* – 9 – 0 have gaps larger than 5%, it is likely that their best-known solutions fall into a local optimum and cannot escape from it because the parameter *prob* is set to 0. As is shown in Table 7, the computation time largely varies, where the computation time in instance *PAP* – 1 – 0 is nearly 10 times longer than the computation time in instance *PAP* – 5 – 0.005. Overall, when parameter *prob* is set to 0.005 or 0.05, the computation time is shorter than others. From these instances, we conclude that an appropriate parameter *prob* is the key to improve the solution efficiency. To balance the efficiency and quality of the solution, we set the parameters *tem* = 3 and *prob* = 0.005 in the following experiments, unless otherwise specified.

5.3. Comparison among different passenger demand scenarios

In this section, instances are implemented under different passenger demand scenarios (i.e., POP, UAT and MHMT). Table 8 reports the detailed experimental results, including the objective function value (OBJ), total waiting time of passengers (TWT), the number of train units used (# TU) and the computation time (CPU). When comparing scenarios UAT and POP, the result shows that when passengers arrive at origin stations uniformly in the temporal dimension, the total waiting time of passengers in the system decreases, but more train units are needed to serve the passengers. While comparing scenarios MHMT and POP, it shows that the number of train units does not change, and the variation of total waiting time can be ignored (with a gap less than 0.2%). In terms of the unified objective function of the three scenarios, all the objective function values are relatively close, which shows that our algorithm can cope with the passenger demand under different scenarios.

Fig. E.1 in Appendix E shows the operation strategies of the three scenarios. In the representation of these timetables, the red solid line indicates that the service is performed by a double-composition, while a blue dashed line indicates a single-composition. In addition, shadows with different colors are used to distinguish sub-lines in the Y-type URT system. Some conclusions can be drawn by comparing the operation strategies of the three scenarios. First, the variation trend of train headways under different scenarios is different. A special example is that in the POP scenario, the headways between downstream services are larger in the early stage of the time horizon than in the late stage,

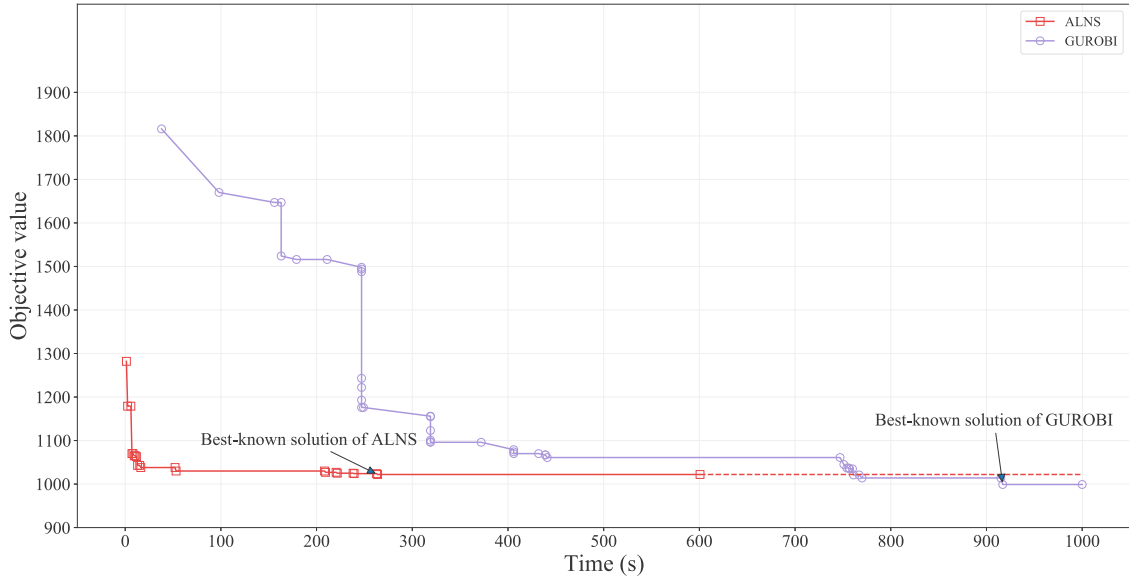


Fig. 9. The convergence tendency of the objective function values.

Table 8

Computational results under different passenger demand scenarios.

Scenarios	OBJ	TWT [min]	# TU	CPU [s]
POP	859 942	229 971	40	1266.29
UAT	881 212	215 606	45	1249.14
MHMT	860 979	230 489	40	2991.53

Table 9

Computational results of different composition modes.

Scenarios	Flexible/Fixed			
	OBJ	TWT [min]	# TU	
POP	859 942/1 200 067	229 971/360 033	40/48	
UAT	881 212/1 187 353	215 606/353 676	45/48	
MHMT	860 979/1 199 515	230 489/359 757	40/48	

while the headways between upstream services show the opposite trend. Second, in the early or late stage of the time horizon, there are more services performed by a single-composition because they only serve the major route. Third, the turnaround time of most train units at turnaround stations is short, which indicates that the utilization efficiency of the train units with flexible train composition mode is high. As a result, we can conclude that our algorithm can formulate effective operation strategies to respond to passenger demand.

5.4. Comparison among different composition modes

To illustrate the potential advantages of the operation plan using flexible train composition mode, we construct a group of instances in which the fixed train composition mode is used. For simplicity, the headway in the instances with fixed train composition mode is set as a constant. In the fixed train composition mode, the services take turns serving the minor route and the major route. To transport more passengers, we assume that all services are performed by a double-composition.

The results for different composition modes are presented in Table 9, where a symbol “/” is used to separate values in the two types of train composition modes. We can see that in each instance, the objective function value of the fixed train composition mode is much larger than the flexible train composition mode, which is mainly affected by the total waiting time.

5.5. Performance comparison between ALNS and an ILP solver

In this section, we compare the performance of ALNS and the general-purpose ILP solver GUROBI in small instances, aiming to verify the effectiveness and efficiency of the former. Please see Appendix F for specific experimental settings and parameters.

As is shown in Table 10, when setting the computation time of GUROBI to 10 min, the optimal solution can be obtained only in the instances with fewer services and passengers. As the number of services and passengers increases, even a feasible solution cannot be obtained. While setting the computation time to three hours, better solutions can be obtained by GUROBI. However, the computation time increases significantly with the increase of the instance scale. In contrast, the increase in computation time of the ALNS algorithm is relatively moderate. In terms of solution quality, the ALNS algorithm also performs well. Specifically, when the time limit of GUROBI is 10 min, the solutions obtained by ALNS are better than GUROBI in most instances, which are marked with * in Table 10. When the time limit of GUROBI is 3 h, the ALNS solutions are still better than GUROBI in some instances, which are marked with □. The last column (Gap) shows that if the ALNS solutions are worse than the GUROBI solutions with a time limit of 3 h, then the relative gaps between them are smaller than 3%. Considering the much shorter computation time, the ALNS significantly outperforms GUROBI in large-scale instances.

We next present the convergence tendency of the objective function values of ALNS and GUROBI. Since the convergence tendency is similar, we only present the results of the instance with 20 services and 500 passengers, as shown in Fig. 9. From these results, we can derive the following observations. (1) As far as the generation time of an initial solution is concerned, that of ALNS is earlier than GUROBI, which represents that our feasibility propositions are effective to obtain a feasible solution quickly. (2) We can see that the solution obtained in close to 50 seconds is good enough in the instance with 20 trains and 500 passengers. We can speculate that ALNS is difficult to find the optimal solution in large-scale instances, but it is more efficient to find a satisfactory solution within an acceptable computation time.

5.6. Sensitivity analysis

In this section, some experiments are performed to verify the rationality of our method.

Table 10

Computational results of comparing GUROBI and ALNS.

# services	# passengers	GUROBI			ALNS		Gap ^c
		OBJ #1 ^a	OBJ #2 ^b	CPU #2 [s]	OBJ	CPU [s]	
16	100	378	378	80.33	386	118.07	2.12%
	300	788	788	2175.85	810	328.92	2.79%
	500	1301	1248	5366.04	1249*	150.34	0.08%
	700	2065	1671	10 800	1682*	412.53	0.66%
	900	None	2112	10 800	2152*	443.49	1.89%
18	100	365	365	210.97	376	101.83	3.01%
	300	725	725	9302.48	750	385.88	3.45%
	500	1171	1113	10 800	1117*	433.58	0.36%
	700	2633	1442	10 800	1442*	579.07	0
	900	None	1950	10 800	1810*	369.83	–
20	100	352	352	347.19	363	322.14	3.13%
	300	696	687	10 800	710	577.45	3.35%
	500	1061	999	10 800	1022*	263.76	2.30%
	700	2082	1351	10 800	1344*	223.37	–
	900	None	2801	10 800	1633*	189.70	–

*The solution of ALNS is not worse than that of GUROBI if solved within a 10-minute time limit.

□ The solution of ALNS is not worse than that of GUROBI if solved within a 3-hour time limit.

^a This column shows the objective function value obtained by GUROBI within a 10-minute time limit of computation.^b This column shows the objective function value obtained by GUROBI within a 3-hour time limit of computation.^c Percentage difference between OBJ and OBJ #2.**Table 11**

Computational results of different objective function weights.

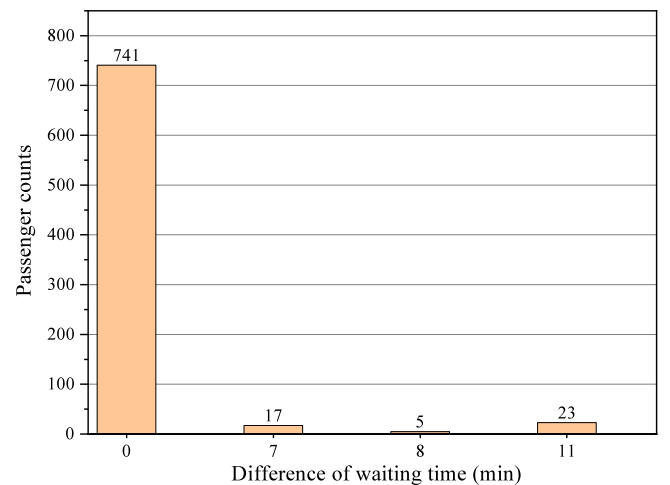
Instance	α	β	OBJ	TWT [min]	# TU	Dominated by ^a	CPU [s]
1	1	0	415 309	207 654	52	–	2302.90
2	1	2000	510 269	209 134	46	–	3217.51
3	1	4000	600 806	212 403	44	–	676.71
4	1	6000	714 718	216 390	47	Instance 2,3	3409.04
5	1	8000	784 950	224 475	42	–	635.34
6	1	10 000	859 942	229 971	40	–	1266.29
7	1	12 000	958 748	227 374	42	Instance 5	3353.72
8	1	14 000	1 042 990	262 495	37	–	2687.27
9	1	16 000	1 087 552	239 776	38	–	3408.15
10	1	18 000	1 180 551	230 275	40	Instance 6	3119.53
11	1	20 000	1 277 893	248 946	39	Instance 9	970.50

^a In this column, ‘–’ means that it is a Pareto optimal solution in these instances; otherwise, it means that the solution is dominated by the solution of the related instances.

First, we compare the waiting time of passengers who travel between the major and minor siding lines with/without pretreatment. In the POP scenario, there are 786 passengers traveling between the major and minor siding lines. Fig. 10 shows the difference in waiting time between two options (i.e., with/without pretreatment). We can see that the difference for most passengers (741/786) is 0, which means that their waiting times are the same regardless of whether there is pretreatment. Considering that these passengers make up a small percentage of all passengers, the impact of our heuristic pretreatment is negligible.

Second, we intend to test the influence of different objective function weights on the best-known solution and computational time, since different weights might correspond to different total waiting times and the number of train units used. To illustrate this effect, we respectively set multiple instances by setting $\alpha = 1$ and changing the value of β in the POP scenario. The detailed computational results are given in Table 11.

In Table 11, we list the values of objective functions and the computational time. Considering that the heuristic algorithm can only obtain a near optimal solution, the experimental results sometimes cannot fully reflect the influence of the weights. Therefore, we eliminate some poor near optimal solutions which are dominated by other solutions, and then the set of Pareto optimal solutions is obtained. By analyzing these Pareto optimal solutions, some variant tendencies are observed easily. With the increase of β , the total waiting time of passengers gradually increases, while the number of train units used gradually decreases.

**Fig. 10.** Computational results of comparing passengers waiting time with/without pretreatment.

The variant trends are reasonable since these two weights in essence represent distinctive decision strategies.

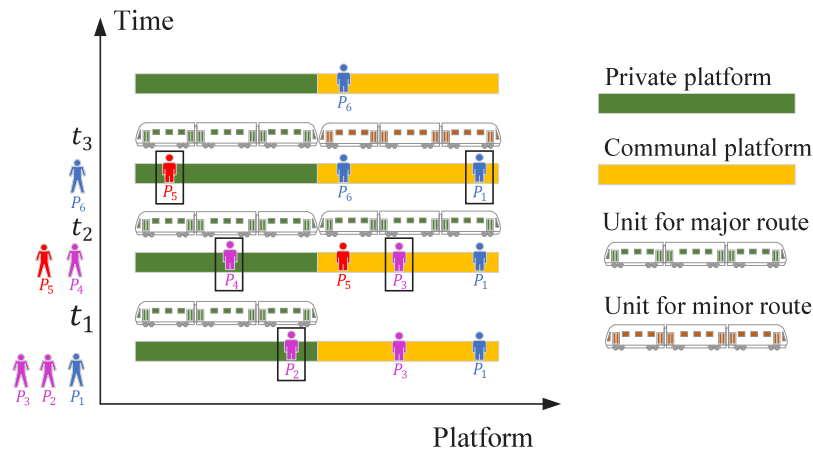


Fig. A.1. An illustration of passenger boarding rules. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table C.1

Values of the parameters.

Parameters	Values
The maximum time stamp, $ T $	480
Dwelling time at station s , t_s^{dt}	30 s
Coupling or decoupling time at joint station, τ^{cd}	30 s
Minimum headway, h^{min}	2 min
Minimum train unit turnaround time, τ^{turn}	2 min
Capacity of one train unit, Cap	900

6. Conclusions and future research

From the perspective of the model, this paper formulates an integrated optimization model of TTP and TUCP for the Y-type URT system to match passenger demand and service capability by using the flexible train composition mode. We first decompose the Y-type system into three sub-lines, and then extend it into a network with joint stations to conveniently model passengers and services on different routes. To deal with the matching problem of train unit routes and passengers' travel paths, the FIFO rules for passengers are specially considered in temporal and spatial dimensions. Besides, by dividing passengers into sets related to passing segments, the constraints for train unit capacity are formulated in a concise form. Compared with the relevant literature, the proposed model is the first time to integrate the train operation plan by considering such a special line (network) and a novel train composition mode in a URT system.

We propose a heuristic algorithm to search a near optimal solution for large-scale real-world cases. Due to the certainty of FIFO rules, we decompose the proposed model into two parts (i.e., TCP and PA, as stated above). To solve the model TCP, we improved the classical ALNS framework by introducing the asynchronous computation mechanism, as well as designing several destroy operators and repair operators according to the characteristics of our problem. For the model PA, a simulation-based method is used to complete the passenger assignments based on a pre-given timetable.

Finally, we perform several groups of computational experiments. Among them, the small-case instances show that the algorithm that we designed is available to obtain near optimal solution to our problem. Meanwhile, in real-world instances, the most important conclusion is that the flexible train composition mode operated on a Y-type system can reduce the passengers' total waiting time, as well as train units used compared with the fixed train composition mode. These results give a managerial insight that the flexible coupling/decoupling activity can better improve service quality, and more appropriately match the passenger demand and the supply of transportation capacity.

There are several interesting directions to extend our problem for future research. First, in our problem, only two types of routes are

considered. However, in a Y-type system, there are at least three types of routes. If considering the combination with the short-turning strategy, more types of routes are involved. Second, more flexible train compositions should be considered. That is to say, future research can focus on coupling or decoupling with more optional train compositions with different numbers of carriages. Third, as more configurations of URT lines emerge, extending our method to other types of lines or networks may be an interesting topic. Fourth, we plan to additionally consider the element of passenger behaviors on the basis of current works, since the passengers are not homogeneous in reality, i.e., one prefers to go to the destination earlier and another prefers to avoid congestion.

CRedit authorship contribution statement

Lin Yang: Writing – original draft, Writing – review & editing, Methodology, Data curation, Formal analysis, Visualization. **Yuan Gao:** Conceptualization, Supervision, Methodology, Writing – review & editing, Funding acquisition. **Andrea D'Ariano:** Writing – review & editing, Methodology, Investigation. **Suxiu Xu:** Writing – review & editing, Visualization.

Declaration of competing interest

All authors declare that they have no conflicts of interest with other researchers and authorities.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 72171023 and 72071093), the Beijing Institute of Technology Research Fund Program for Young Scholars, China, and the RGC TRS Project (T32-707-22-N). Besides, we appreciate the editors and the anonymous reviewers for their valuable comments and recommendations.

Appendix A. An example to illustrate the boarding rules

The three types of passengers are represented by three colors: trunk passengers are represented by red, major route passengers are represented by pink, and minor route passengers are represented by blue. The platform of a trunk line station is divided into two parts: communal platform and private platform. All train units can stop beside

Table C.2
Running time in segments.

Segments	Running time (unit: 30 s)	Segments	Running time (unit: 30 s)	Segments	Running time (unit: 30 s)
#JHWG - BYDP	7	XH* - TP	9	XN - FX	3
BYDP - XL	5	TP - SG	7	FX - ZS	5
XL - TH	7	SG - CC	9	ZSC - HTX	5
TH - ZL	9	CC - CHKYZ	9	HTX - WC	3
ZL - ZLT	7	CHKYZ - #DF	5	WC - TC	5
ZLT - ML	9	XH* - HW	5	TC - ZLB	5
ML - XH*	5	HW - XN	5	ZLB - #ZL	3

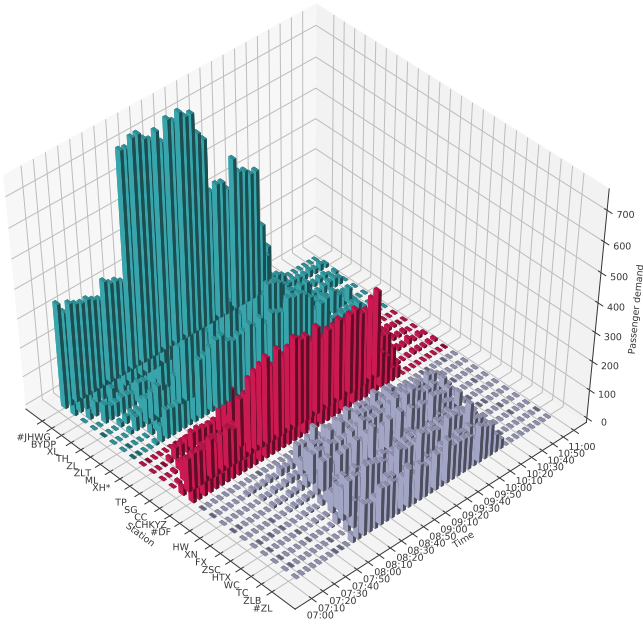


Fig. D.1. An illustration of the passenger demand under the POP scenario.

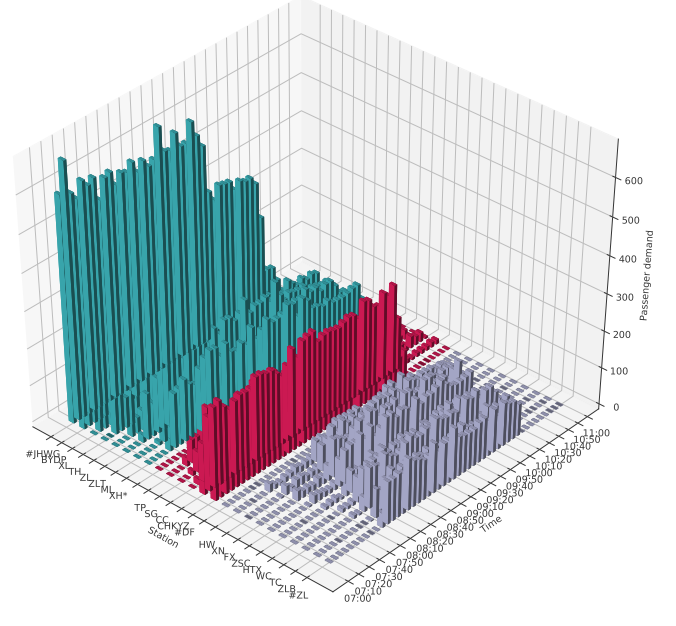


Fig. D.2. An illustration of the passenger demand under the UAT scenario.

the communal platform, while only train units associated with the major route can stop beside the private platform. That is to say, minor route passengers can only wait at the communal platform to board train units. As shown in Fig. A.1, all minor route passengers (marked with blue) wait at the communal platform, while the major route passengers (marked with pink) and trunk passengers (marked with red) can wait at both platforms.

In Fig. A.1, six downstream passengers are numbered by their arrival sequence. To illustrate the rules for passengers boarding train units, assume that there is only one spare seat in each train unit when arriving at the station. If a passenger marked with black wireframe indicates that it is his/her turn to board the train unit of the next service. In this paper, passengers boarding the train unit follow the principle of route-based FIFO, which is divided into temporal and spatial dimensions. The temporal FIFO represents that for some passengers who want to board the same train unit at a station, then the passenger who arrives earlier has higher priority. For example, passenger P_2 and passenger P_3 are major route passengers, and both of them want to board service 1. However, there is only one spare seat in the train unit of service 1, so the earlier passenger P_2 can board it, while passenger P_3 has to wait for the next service. The temporal FIFO does not apply to passengers who want to board different train units, which can be explained by passenger P_1 and passenger P_3 , where passenger P_1 arrives earlier but board the train unit later. The rule called spatial FIFO means that a passenger whose turn it is must board the train unit. As shown in Fig. A.1, all passengers marked with black wireframe board the train unit of the next service instead of waiting for another service any longer.

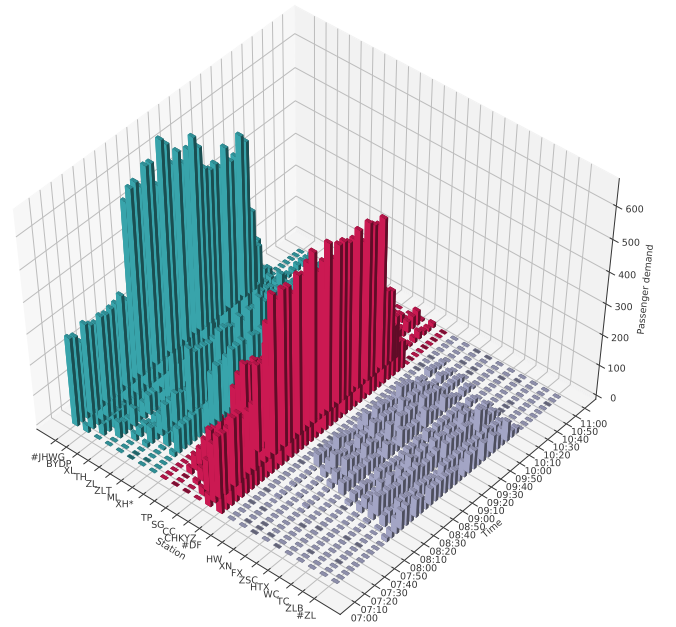
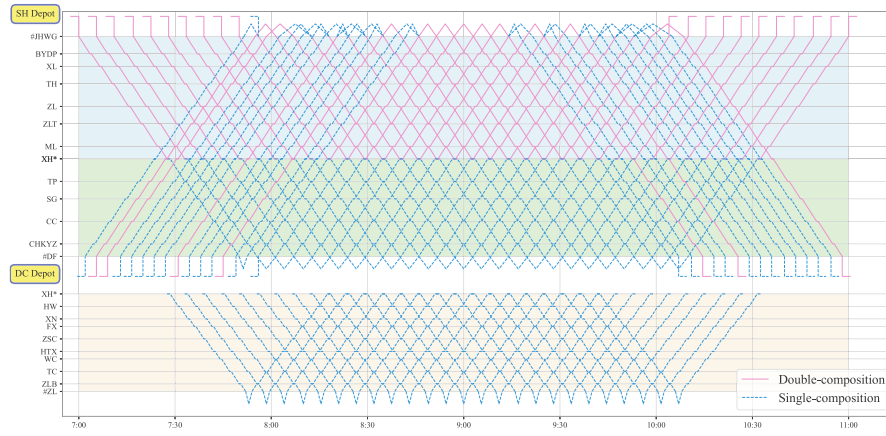


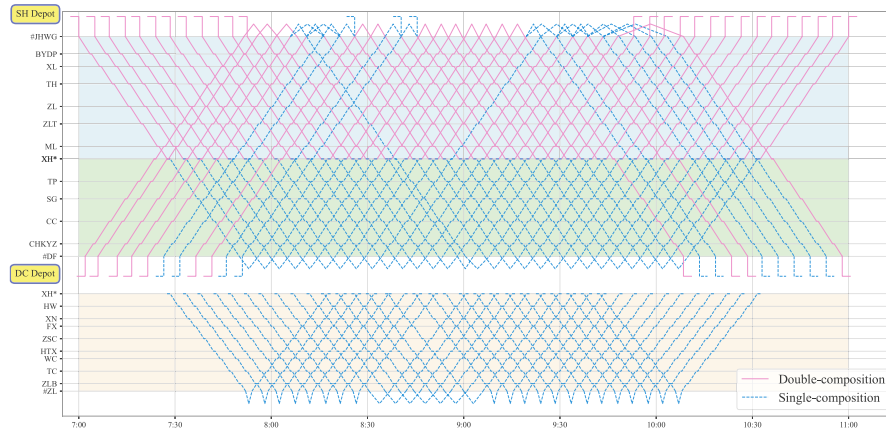
Fig. D.3. An illustration of the passenger demand under the MHMT scenario.

Appendix B. Linearization of the model

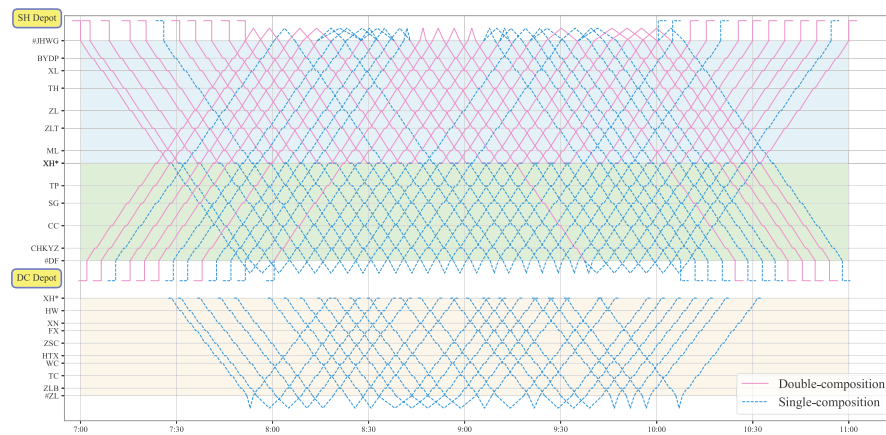
Note that there is a term in Constraint (27) that multiplies two variables, so it is a quadratic objective function rather than a linear



(a) POP



(b) UAT



(c) MHMT

Fig. E.1. The operation strategies of different passenger demand scenarios. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

one. To solve our problem by an ILP solver in small-case instances, we linearize the non-linear expressions. Wu et al. [27] handled a similar problem. The linearization method for our problem can be summarized as follows,

Step 1. Introduce a set of auxiliary variables $\omega_p (p \in P)$, which means the waiting time of passenger p . The following two constraints hold,

$$\omega_p \geq 0, \quad \forall p \in P \quad (\text{B.1})$$

$$\omega_p \geq M \cdot (z_{p,i,r} - 1) + t_{i,o_p}^{dep} - t_p, \quad \forall p \in P, i \in I, r \in R \quad (\text{B.2})$$

Step 2. Reformulate the objective of the total waiting time of all passengers,

$$\min \sum_{p \in P} \omega_p \quad (\text{B.3})$$

Constraint (B.1) means that the waiting time of passenger p is nonnegative, which is a matter of course. In Constraint (B.2), only if $z_{p,i,r} = 1$, it gives a reliable lower bound $t_{i,o_p}^{dep} - t_p$ of ω_p , otherwise it is invalid because it is slacker than Constraint (B.1). Finally, the minimum total waiting time of all passengers can be reformulated as Expression (B.3), and then the integer linear programming (ILP) model can be reformulated as Expression (B.4).

$$\min \quad \alpha \cdot \sum_{p \in P} \omega_p + \beta \cdot \sum_{i \in I} \sum_{s \in S^s} o_{i,s} \quad (\text{B.4})$$

s.t. Constraints (1)–(26), (B.1), and (B.2)

Appendix C. The parameters of experiments on real-world instances

See Tables C.1 and C.2

Appendix D. Passenger demand under different scenarios

In this appendix, we show the passenger demand arriving at stations in different scenarios. Here are the specific scenarios and the characteristics of the passenger demands (see Figs. D.1–D.3).

POP: The passenger demand with peak period and off-peak period.

UAT: The passenger demand with uniform density in the temporal dimension.

MHMT: The passenger demand by reducing the density of passenger demand on the minor siding line by half and increasing in the density of passenger demand on the major siding line to twice based on the POP scenario.

Appendix E. The operation strategies

In this appendix, the operation strategies corresponding to the three scenarios in Appendix D are shown.

Appendix F. Experimental settings of small-case instances

Assume a Y-type system with three stations on each sub-line and no depot on the minor siding line, and the time horizon is discretized to 50 by a time slot with a 1-minute length. The dwelling time at stations t_s^{dt} , and the coupling or decoupling time τ^{cd} are all set to 1 min. The minimum headway h^{min} and minimum turnaround time τ^{turn} are set to 2 min. The capacity of one train unit is set to 50. The objective function weights α, β take the value of 1 and 20, respectively.

We generate a total of 15 instances by changing the number of services and passengers to test the performance of the ALNS algorithm and GUROBI. In these instances, we gradually increase the number of services (i.e., 16, 18 and 20) and the number of passengers (i.e., from 100 to 900 in increments of 200). Considering that when we solve the mathematical programming problem by a commercial solver, most of

the time it is actually used to prove the optimality of the obtained solution instead of updating a new solution. Therefore, in our small-case instances, we set the computation time limit of each experiment solved by GUROBI to 3 h to obtain a near optimal solution. When using the ALNS algorithm, we set the maximum iteration counts to 1000, the maximum computing time to 10 min, and the maximum time to search for the next better solution to 3 min.

References

- [1] Aoun J, Quaglietta E, Goverde RM. Investigating market potentials and operational scenarios of virtual coupling railway signaling. *Transp Res Rec* 2020;2674(8):799–812.
- [2] Haltuf M. Shift2Rail JU from member state's point of view. *Transp Res Procedia* 2016;14:1819–28.
- [3] Pan H, Yang L, Liang Z. Demand-oriented integration optimization of train timetabling and rolling stock circulation planning with flexible train compositions: A column-generation-based approach. *European J Oper Res* 2022.
- [4] Zhou H, Qi J, Yang L, Shi J, Pan H, Gao Y. Joint optimization of train timetabling and rolling stock circulation planning: A novel flexible train composition mode. *Transp Res B* 2022;162:352–85.
- [5] Ying C-s, Chow AH, Nguyen HT, Chin K-S. Multi-agent deep reinforcement learning for adaptive coordinated metro service operations with flexible train composition. *Transp Res B* 2022;161:36–59.
- [6] Bussieck MR, Winter T, Zimmermann UT. Discrete optimization in public rail transport. *Math Program* 1997;79(1):415–44.
- [7] Gao Y, Schmidt M, Yang L, Gao Z. A branch-and-price approach for trip sequence planning of high-speed train units. *Omega* 2020;92:102150.
- [8] Cacchiani V, Huisman D, Kidd M, Kroon L, Toth P, Veelenturf L, Wagenaar J. An overview of recovery models and algorithms for real-time railway rescheduling. *Transp Res B* 2014;63:15–37.
- [9] Lamorgese L, Mannino C. An exact decomposition approach for the real-time train dispatching problem. *Oper Res* 2015;63(1):48–64.
- [10] Lamorgese L, Mannino C, Piacentini M. Optimal train dispatching by Benders'-like reformulation. *Transp Sci* 2016;50(3):910–25.
- [11] Lamorgese L, Mannino C, Natvig E. An exact micro-macro approach to cyclic and non-cyclic train timetabling. *Omega* 2017;72:59–70.
- [12] Mannino C, Nakkerud A. Optimal train rescheduling in oslo central station. *Omega* 2023;116:102796.
- [13] Samà M, Meloni C, D'Ariano A, Corman F. A multi-criteria decision support methodology for real-time train scheduling. *J Rail Transp Plan Manag* 2015;5(3):146–62.
- [14] Yin J, Tang T, Yang L, Gao Z, Ran B. Energy-efficient metro train rescheduling with uncertain time-variant passenger demands: An approximate dynamic programming approach. *Transp Res B* 2016;91:178–210.
- [15] Zhu Y, Goverde RM. Dynamic and robust timetable rescheduling for uncertain railway disruptions. *J Rail Transp Plan Manag* 2020;15:100196.
- [16] Zhu Y, Goverde RM. Railway timetable rescheduling with flexible stopping and flexible short-turning during disruptions. *Transp Res B* 2019;123:149–81.
- [17] Zhu Y, Goverde RM. Integrated timetable rescheduling and passenger reassignment during railway disruptions. *Transp Res B* 2020;140:282–314.
- [18] Zhu Y, Goverde RM. Dynamic railway timetable rescheduling for multiple connected disruptions. *Transp Res C* 2021;125:103080.
- [19] Vuchic VR. *Urban transit: Operations, planning, and economics*. John Wiley & Sons; 2017.
- [20] Niu H, Zhou X. Optimizing urban rail timetable under time-dependent demand and oversaturated conditions. *Transp Res C* 2013;36:212–30.
- [21] Niu H, Zhou X, Gao R. Train scheduling for minimizing passenger waiting time with time-dependent demand and skip-stop patterns: Nonlinear integer programming models with linear constraints. *Transp Res B* 2015;76:117–35.
- [22] Yang L, Qi J, Li S, Gao Y. Collaborative optimization for train scheduling and train stop planning on high-speed railways. *Omega* 2016;64:57–76.
- [23] Gao Y, Kroon L, Yang L, Gao Z. Three-stage optimization method for the problem of scheduling additional trains on a high-speed rail corridor. *Omega* 2018;80:175–91.
- [24] Zhang C, Gao Y, Yang L, Kumar U, Gao Z. Integrated optimization of train scheduling and maintenance planning on high-speed railway corridors. *Omega* 2019;87:86–104.
- [25] Cacchiani V, Qi J, Yang L. Robust optimization models for integrated train stop planning and timetabling with passenger demand uncertainty. *Transp Res B* 2020;136:1–29.
- [26] Chen Z, Li S, D'Ariano A, Yang L. Real-time optimization for train regulation and stop-skipping adjustment strategy of urban rail transit lines. *Omega* 2022;110:102631.
- [27] Wu Y, Yang H, Zhao S, Shang P. Mitigating unfairness in urban rail transit operation: A mixed-integer linear programming approach. *Transp Res B* 2021;149:418–42.

- [28] Tang L, D'Ariano A, Xu X, Li Y, Ding X, Samà M. Scheduling local and express trains in suburban rail transit lines: Mixed-integer nonlinear programming and adaptive genetic algorithm. *Comput Oper Res* 2021;135:105436.
- [29] Schrijver A. Minimum circulation of railway stock. *Cwi Q* 1993;6(3):205–17.
- [30] Lusby RM, Haahr JT, Larsen J, Pisinger D. A branch-and-price algorithm for railway rolling stock rescheduling. *Transp Res B* 2017;99:228–50.
- [31] Steinzen I, Gintner V, Suhl L, Kliewer N. A time-space network approach for the integrated vehicle-and crew-scheduling problem with multiple depots. *Transp Sci* 2010;44(3):367–82.
- [32] Cacchiani V, Caprara A, Toth P. An effective peak period heuristic for railway rolling stock planning. *Transp Sci* 2019;53(3):746–62.
- [33] Abbink E, Van den Berg B, Kroon L, Salomon M. Allocation of railway rolling stock for passenger trains. *Transp Sci* 2004;38(1):33–41.
- [34] Canca D, Barrena E. The integrated rolling stock circulation and depot location problem in railway rapid transit systems. *Transp Res E* 2018;109:115–38.
- [35] Alfieri A, Groot R, Kroon L, Schrijver A. Efficient circulation of railway rolling stock. *Transp Sci* 2006;40(3):378–91.
- [36] Peeters M, Kroon L. Circulation of railway rolling stock: a branch-and-price approach. *Comput Oper Res* 2008;35(2):538–56.
- [37] Fioole P-J, Kroon L, Maróti G, Schrijver A. A rolling stock circulation model for combining and splitting of passenger trains. *European J Oper Res* 2006;174(2):1281–97.
- [38] Lin Z, Kwan RS. A branch-and-price approach for solving the train unit scheduling problem. *Transp Res B* 2016;94:97–120.
- [39] Cacchiani V, Caprara A, Toth P. Solving a real-world train-unit assignment problem. *Math Program* 2010;124(1):207–31.
- [40] Cacchiani V, Caprara A, Toth P. A Lagrangian heuristic for a train-unit assignment problem. *Discrete Appl Math* 2013;161(12):1707–18.
- [41] Thorlacius P, Larsen J, Laumanns M. An integrated rolling stock planning model for the Copenhagen suburban passenger railway. *J Rail Transp Plan Manag* 2015;5(4):240–62.
- [42] Hoogervorst R, Dollevoet T, Maróti G, Huisman D. A Variable Neighborhood Search heuristic for rolling stock rescheduling. *EURO J Transp Logist* 2021;10:100032.
- [43] Wang Y, Tang T, Ning B, Meng L. Integrated optimization of regular train schedule and train circulation plan for urban rail transit lines. *Transp Res E* 2017;105:83–104.
- [44] Yuan J, Gao Y, Li S, Liu P, Yang L. Integrated optimization of train timetable, rolling stock assignment and short-turning strategy for a metro line. *European J Oper Res* 2021.
- [45] Wang D, D'Ariano A, Zhao J, Zhong Q, Peng Q. Integrated rolling stock deadhead routing and timetabling in urban rail transit lines. *European J Oper Res* 2022;298(2):526–59.
- [46] Wang Y, Zhao K, D'Ariano A, Niu R, Li S, Luan X. Real-time integrated train rescheduling and rolling stock circulation planning for a metro line under disruptions. *Transp Res B* 2021;152:87–117.
- [47] Liu R, Li S, Yang L. Collaborative optimization for metro train scheduling and train connections combined with passenger flow control strategy. *Omega* 2020;90:101990.
- [48] Cadarso L, Marín Á. Integration of timetable planning and rolling stock in rapid transit networks. *Ann Oper Res* 2012;199(1):113–35.
- [49] Cadarso L, Marín Á, Maróti G. Recovery of disruptions in rapid transit networks. *Transp Res E* 2013;53:15–33.
- [50] Zhao Y, Li D, Yin Y, Zhao X. Integrated optimization of demand-driven timetable, train formation plan and rolling stock circulation with variable running times and dwell times. *Transp Res E* 2023;171:103035.
- [51] Yang A, Wang B, Huang J, Li C. Service replanning in urban rail transit networks: Cross-line express trains for reducing the number of passenger transfers and travel time. *Transp Res C* 2020;115:102629.
- [52] Wang Y, Zhu S, D'Ariano A, Yin J, Miao J, Meng L. Energy-efficient timetabling and rolling stock circulation planning based on automatic train operation levels for metro lines. *Transp Res C* 2021;129:103209.
- [53] Pisinger D, Ropke S. A general heuristic for vehicle routing problems. *Comput Oper Res* 2007;34(8):2403–35.
- [54] Sacramento D, Pisinger D, Ropke S. An adaptive large neighborhood search metaheuristic for the vehicle routing problem with drones. *Transp Res C* 2019;102:289–315.
- [55] Gu W, Cattaruzza D, Ogier M, Semet F. Adaptive large neighborhood search for the commodity constrained split delivery VRP. *Comput Oper Res* 2019;112:104761.
- [56] Santos MJ, Jorge D, Ramos T, Barbosa-Póvoa A. Green reverse logistics: Exploring the vehicle routing problem with deliveries and pickups. *Omega* 2023;118:102864.
- [57] Real LB, Contreras I, Cordeau J-F, de Camargo RS, de Miranda G. Multimodal hub network design with flexible routes. *Transp Res E* 2021;146:102188.
- [58] Mohri SS, Karimi H, Kordani AA, Nasrollahi M. Airline hub-and-spoke network design based on airport capacity envelope curve: a practical view. *Comput Ind Eng* 2018;125:375–93.
- [59] He K, Tole K, Ni F, Yuan Y, Liao L. Adaptive large neighborhood search for solving the circle bin packing problem. *Comput Oper Res* 2021;127:105140.
- [60] Li Y, Chen M, Huo J. A hybrid adaptive large neighborhood search algorithm for the large-scale heterogeneous container loading problem. *Expert Syst Appl* 2022;189:115909.
- [61] Yin J, D'Ariano A, Wang Y, Yang L, Tang T. Timetable coordination in a rail transit network with time-dependent passenger demand. *European J Oper Res* 2021;295(1):183–202.
- [62] Barrena E, Canca D, Coelho LC, Laporte G. Single-line rail rapid transit timetabling under dynamic passenger demand. *Transp Res B* 2014;70:134–50.
- [63] Ropke S, Pisinger D. An adaptive large neighborhood search heuristic for the pickup and delivery problem with time windows. *Transp Sci* 2006;40(4):455–72.